

Should Central Banks Issue Digital Currency with a Cap? *

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Abstract

I study how introducing a central bank digital currency with a holding limit affects equilibrium allocations and welfare in an environment where a digital currency competes with bank deposits. Imposing a limit on households' digital currency holdings may not be optimal, depending on other design features. I show that when the fixed cost and the interest rate on the digital currency are set properly, it is optimal to impose no limit. However, imposing a limit improves welfare when the central bank fails to set the interest rate optimally.

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1 Introduction

With advancements in mobile payment technology, the growth of online retailing, and shocks such as the COVID-19 pandemic, the decline in cash use by the public has become both inevitable and pronounced. This declining demand for cash as a means of payment is matched by a rising demand for digital monies—such as bank deposits, digital wallets, and stablecoins—provided by the private sector. However, the dominant role of private monies in the payment landscape can have some negative consequences. For example, it can hinder financial inclusion—disadvantaging the unbanked and underbanked—and increase the payment system’s vulnerability during times of disruption.¹ These concerns, along with other considerations such as supporting monetary sovereignty and expanding the central bank’s monetary policy toolkit, have prompted central bankers to consider issuing retail central bank digital currencies (CBDCs).² A retail CBDC is an electronic form of central bank money that is accessible to the general public for retail transactions.³

Since CBDC is a new payment solution, its design can have important implications for existing payment service providers, particularly private banks. A CBDC can crowd out bank deposits by raising the funding costs for private banks. Consequently, central banks aim to design CBDCs that lead to minimal financial disruption. For example, Fabio Panetta, then a member of the Executive Board of the European Central Bank, emphasized “we will make sure that the digital euro...does not crowd out private means of payment, and preserves financial stability” (Panetta, 2022). One design feature proposed by central bankers to help retain bank deposits is imposing a cap on the amount of CBDC that households can hold or use. More than half of the 93 central banks surveyed by the Bank for International Settlements (BIS) in 2024 consider a (potential) CBDC that is subject to holding limits (see Anamaria Illes and Wierds, 2025).⁴ Moreover, Li et al. (2024) find that a large holding limit

¹The legal tender status of cash means that cash must be accepted for debt repayment. However, retail transactions are not always considered as debt repayment. They are private agreements between buyers and sellers. In many countries, including Sweden, the UK, and the US, businesses can refuse cash if they state their policy upfront. Businesses may increasingly opt to do so if cash is less often used in an attempt to streamline the transaction process.

²For example, a primary motivation for the ECB to issue a digital euro is to provide Europeans with a payment rail based on European technology and infrastructure—a critical objective amidst current geopolitical turbulence. As of July 2025, 65 countries are exploring a retail CBDC, according to Atlantic Council, a think tank. Three countries have issued their own retail CBDCs, which are the Bahamas, Jamaica and Nigeria.

³I use the term CBDC to refer to a retail CBDC in the subsequent analysis.

⁴For instance, the European Central Bank has considered a €3,000 to €4,000 limit per individual for digital euro holdings (Panetta, 2022). The Bank of England has suggested a per capita cap between £10,000 and £20,000 for the digital pound (Cunliffe, 2023), while the majority of commercial bank respondents in a 2023 consultation survey favoured lower limits between £3,000 and £5,000 (see Bank of England and HM Treasury, 2024). Meanwhile, the People’s Bank of China has introduced multiple limits for the digital yuan,

of 25,000 Canadian digital dollars can substantially mitigate the crowding out without a large effect on depositors' welfare using a heterogeneous-agent framework.

I study the optimal design of a CBDC that competes with bank deposits as a medium of exchange, using a New Monetarist model in the tradition of Lagos and Wright (2005). On the one hand, introducing a CBDC can improve efficiency in exchange by paying a higher interest rate than deposits; on the other hand, it can crowd out bank-funded investment—the same tradeoff as in Keister and Sanches (2022). I show that introducing a CBDC can often lead to a higher welfare in this environment, despite the disintermediation effect, when the central bank gains three new policy tools: the CBDC interest rate, holding limit and fixed cost. The optimal design of a CBDC is sensitive to which existing form(s) of payment it competes with. I study two possibilities, one in which the central bank issues a universal CBDC, which competes with both physical currency and deposits, and the other in which I have a deposit-like CBDC, which only competes with bank deposits.

I begin the analysis of a universal CBDC. A high interest rate and a binding CBDC holding limit can allow the CBDC to coexist with cash and bank deposits at the intensive margin, and furthermore, a CBDC fixed cost allows the CBDC to coexist with bank deposits also at the extensive margin. Therefore, we can have different equilibria under different CBDC designs. In this case, the optimal limit is often nonbinding, partly due to the large increase in decentralized trade facilitated via cash before a CBDC is introduced, as the limit increases. And this motivates me to look at the optimal design of a deposit-like CBDC.

The analysis of a deposit-like CBDC can provide supporting evidence for having a binding holding limit. I show that when the fixed cost and the interest rate on the CBDC are set properly, it is optimal to impose no limit. However, imposing a limit improves welfare under some other designs, especially when the central bank fails to set the optimal interest rate. In such cases, the optimal limit lies within the regime where the CBDC coexists with bank deposits both at the intensive and extensive margins. Bidder et al. (2025) also suggests that introducing a CBDC with a holding limit improves welfare under some calibrations of a DSGE model.⁵

Related literature. This paper is mainly related to two lines of the literature on CBDC. The first line assesses the macroeconomic effects of introducing a CBDC, in particular its

including a single payment limit, a daily cumulative limit, and a balance cap, which vary depending on the level of anonymity the digital wallet provides (Mu, 2022). For example, the most anonymous level-four wallet has a single transaction limit of 2,000 RMB (approximately 276 USD). This is calculated based on the Q2 2025 average USD/RMB exchange rate provided by FRED.

⁵In their paper, a CBDC can lead to disintermediation both in normal times and in distress, where disintermediation in normal times has positive effect on welfare because it reduces the size of the banking sector which is prone to runs, whereas disintermediation in distress deteriorates welfare.

effect on private bank intermediation.⁶ Andolfatto (2021) examines the impact of a CBDC on a monopolistic banking sector within an overlapping generations framework. Williamson (2022b) shows that introducing a CBDC as a narrow banking facility—which competes with private banks subject to incentive problems—can increase welfare. Chiu et al. (2023) shows that a CBDC can crowd in deposits when banks have market power. In Chiu and Davoodalhosseini (2023), banks issue deposits to finance investments that produce goods traded in the decentralized market, and this general equilibrium feedback effect generates the result that an interest-bearing, cash-like CBDC can crowd in bank deposits when the banking sector is perfectly competitive. Dong and Xiao (2024) obtain similar results with a different mechanism where CBDC and bank deposits become complements. In their model, entrepreneurs without investment opportunities deposit their idle CBDC at banks, which are then used as reserves to issue loans. Within this line of literature, my paper builds on Keister and Sanches (2022). In Keister and Sanches (2022), disintermediating banks could be costly because of banks’ limited commitment, which can make investment inefficiently low in equilibrium, and they use the interest payment on CBDC as a policy tool to balance this disintermediation effect against the improvements in exchange efficiency. In this paper, I introduced the CBDC holding limit as an additional tool to balance this tradeoff, and the fixed costs for using deposits and the CBDC make the welfare analysis and thereby the optimal design less straightforward.

The second line is the literature on the optimal design of a CBDC, where privacy is mostly discussed. Agur et al. (2022) investigate the optimal level of anonymity a CBDC should provide in the presence of network effects. Similarly, Wang (2023) examines the optimal privacy design of a CBDC, but in the context of money laundering. This paper abstracts from the issue of privacy, and it is implicitly assumed that a CBDC can be as anonymous as cash while also as transparent as deposits.

2 The Model

In this section, I paint the physical environment, which combines features of Lagos and Wright (2005) and Keister and Sanches (2022). I also characterize and define equilibrium when agents have access to a general portfolio of monies. Subsequent sections then specialize the discussions on different CBDC designs.

⁶Jiang and Zhu (2021) discuss the policy interaction between the interest on a CBDC and the interest on reserves, Barrdear and Kumhof (2022) use a quantitative DSGE model to evaluate the macroeconomic consequences of a CBDC, Davoodalhosseini (2022) examines the monetary policy implications of a CBDC with balance-contingent transfers.

2.1 The environment

Time is discrete and with infinite horizon. Each period is divided into two subperiods: a frictional decentralized market (DM) and a Walrasian centralized market (CM). There are two perishable goods which are produced and consumed in the two subperiods respectively: the DM good and the CM good.

Agents. There are three types of private agents: a unit measure each of buyers and sellers, and a continuum of bankers. Buyers and sellers live forever and discount across periods with factor $\beta \in (0, 1)$. In the DM, the first subperiod, buyers and sellers meet bilaterally, where buyers consume what sellers produce and not vice versa. By consuming q units of the DM good, a buyer's utility is $u(q)$ with $u(0) = 0, u'(0) = \infty, u' > 0$, and $u'' < 0$. Sellers incur a linear cost in producing q units of the DM good with $c(q) = q$. Hence, the efficient amount of DM trade q^* , the one maximizing the total trade surplus, solves $u'(q^*) = 1$. Then in the second subperiod, the CM, both buyers and sellers can work and consume the CM good. By working h hours, they produce h units of the CM good with disutility $-h$. Their preferences for CM consumption is $U(\cdot)$ with $U'(0) = \infty, U' > 0$, and $U'' < 0$. Hence, the equilibrium CM consumption X^* , solves $U'(X^*) = 1$. In summary, buyers' and sellers' period utilities are

$$\begin{aligned} U^b(q, X, h) &= u(q) + U(X) - h, \\ U^s(q, X, h) &= -q + U(X) - h, \end{aligned}$$

where X is consumption of the CM good.

There are overlapping generations of bankers and a new cohort of bankers is born each CM. They become old the next period and die at the end of next CM. They are active only in the CM and only consume when they are old. Young bankers are endowed with an indivisible investment project but they have no internal funds to finance it. Therefore, they must borrow from households by issuing deposits. Investment projects take one unit of current CM good as input and have heterogeneous returns in the next CM. There is a measure η of bankers with each return in the support $[0, \bar{\gamma}]$, which are known in advance. Bankers have limited commitment and therefore their project returns are not fully pledgeable. Young bankers can credibly pledge only a fraction $\epsilon \in [0, 1]$ of their project returns to depositors, as they can abscond with the remainder. Moreover, bankers have to satisfy a reserve requirement of μ for each unit of deposits issued.

Assets and exchange. Agents other than the central bank lack commitment and there is no record-keeping technology among buyers and sellers so DM trade must be *quid pro quo*. Buyers will have to use means of payment to purchase and consume DM goods. Furthermore, households are perfectly and permanently segmented into two types of meetings where different means of payment can be used for transactions in the DM before they make their portfolio choices in the CM: a fraction λ_1 of households engages in Type 1 meetings, while the remaining fraction $\lambda_2 \equiv 1 - \lambda_1$ participates in Type 2 meetings.⁷ Before a CBDC is introduced, households can use only cash in Type 1 meetings and only deposits in Type 2 meetings. We can think of Type 1 meetings as point-of-sale transactions where sellers do not have the technology to verify bank deposits, and Type 2 meetings as transactions where sellers cannot verify physical currency. When a CBDC is introduced, two different kinds of designs are discussed—one that competes only with deposits and another that competes with both physical currency and deposits. The former is referred to as a deposit-like CBDC and the latter a universal CBDC. I abstract from the case of a cash-like CBDC, given that a digital currency is inherently unlikely to compete solely with physical cash. The three possible media of exchange are embedded with different characteristics. First, they have different rates of return. Returns on cash and the CBDC are controlled by the central bank through targeting the inflation rate and the interest payment on the CBDC whereas the deposit rate is an equilibrium object. Second, the real balances of the CBDC households can hold per capita are subject to a cap, denoted by $\bar{e}$, which can be binding or nonbinding.⁸ Third, buyers incur strictly positive fixed costs for using deposits and the CBDC. The CBDC fixed cost δ consists of two components: a user cost δ_1 and a subsidy (service charge) when $\delta_2 < (>)0$, where δ_2 is a policy tool.

The flow of payments and goods is as follows. In the CM, buyers in both types of meetings make their portfolio choices, i.e., which means of payment and how much to bring into the next DM for trade. When bank deposits and the CBDC are accumulated, fixed costs are payable at this stage. Young bankers issue deposits to finance their investment. In the following DM, decentralized trade takes place, where money and goods change hands. In the next CM, sellers redeem deposits received with now old bankers and consume. Old bankers

⁷Results are the same when the segmentation is random each period as long as the proportions of households engaging in Type 1 and Type 2 meetings remain the same.

⁸For simplicity, I assume that households hold their CBDC accounts directly at the central bank, allowing the central bank to monitor and regulate households' CBDC balances. If, instead, households were required to hold CBDC accounts at commercial banks, I assume that it would be technologically feasible for the central bank to track and monitor households' balances across different banks. In addition, I impose a real holding limit, whereas central banks have typically communicated nominal limits. In stationary equilibria, the price of money changes across periods, and thus the nominal CBDC holding limit would also need to change in order to keep the real balances of CBDC constant. Therefore, using a real CBDC holding limit is more suitable for steady-state analysis.

consume what is left after repaying deposits with interests. A new cohort of young bankers is born endowed with investment projects. Buyers receive lump-sum transfers and adjust their balances. Figure 1 summarizes activities carried out by private agents in a timeline.

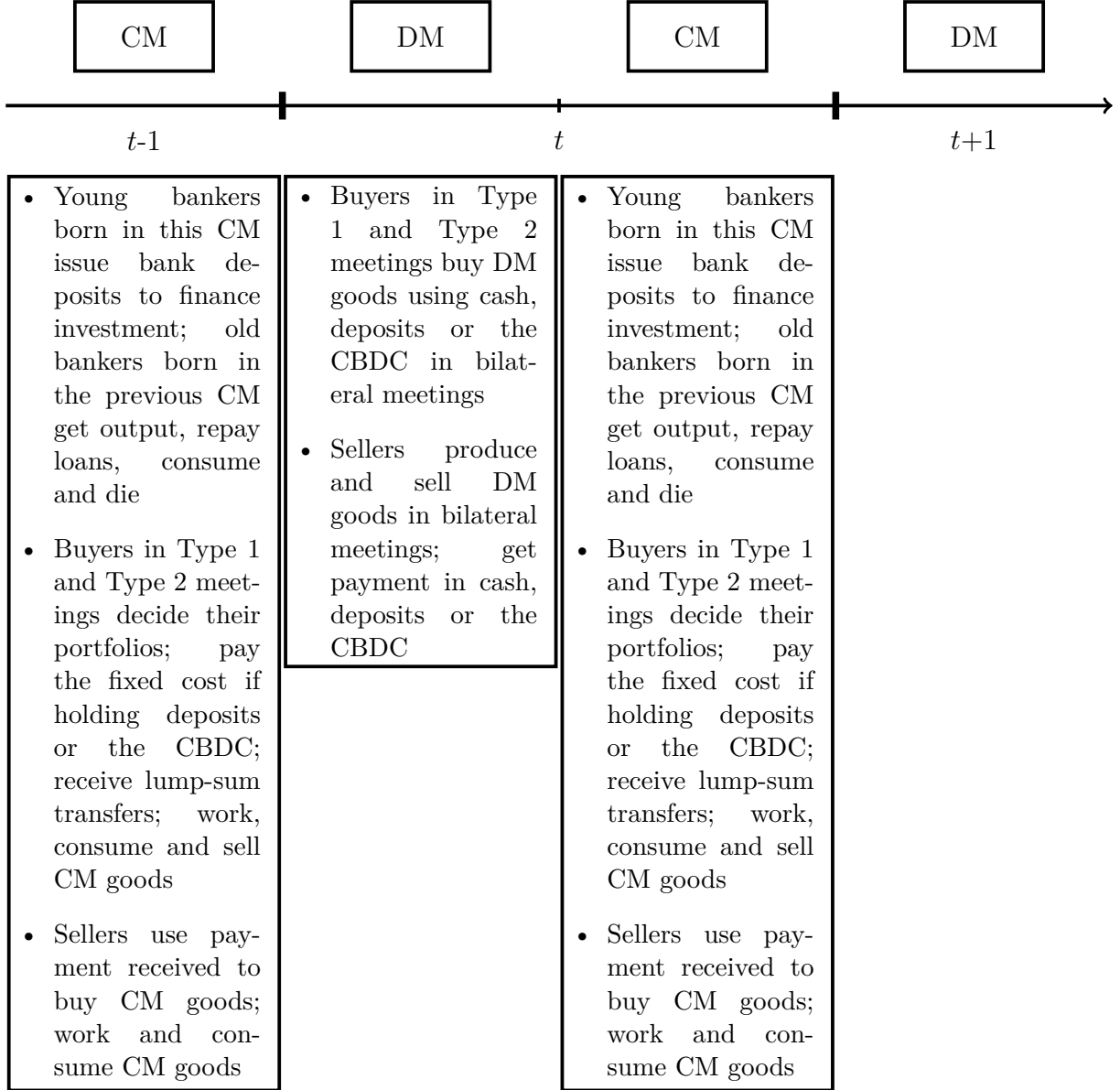


Figure 1: Timeline.

2.2 Asset demand

In this section, I solve buyers' portfolio problems in both Type 1 and Type 2 meetings within a generalized setup to derive households' demand for the three means of payment: cash, deposits and the CBDC.

In what follows, time subscripts on period- t variables are omitted, while variables from $t - 1$ are labeled with subscript -1 , and variables from $t + 1$ with subscript $+1$. Denote the price of money in terms of the CM good in period t as ϕ . Define $\vec{a} = (c, d, e)$ as the portfolio vector of real balances of cash, deposits and CBDC accumulated by a buyer. The net nominal interest rate on CBDC balances is i . Let $\vec{R} = (R^c, R^d, R^e)$ be the vector of real gross returns on cash, deposits and the CBDC carried from period $t - 1$ to period t , where $R^c = \frac{\phi}{\phi_{-1}}$, $R^e = \frac{(1+i)\phi}{\phi_{-1}}$, and R^d is determined in equilibrium. Here it is implicitly assumed that there is one-to-one exchange rate between cash and the CBDC, as they have the same price ϕ .

Bellman equations. Let $W_j(\vec{a})$ denote the value function for a buyer entering the CM in period t with portfolio $\vec{a}$ who participates in Type $j \in \{1, 2\}$ meetings. Let $V_j(\vec{a}_{+1})$ denote for the same buyer, the value function for entering the DM with portfolio $\vec{a}_{+1}$.

In the CM, a buyer chooses her consumption of the CM good X^b , labor h , and portfolio $\vec{a}_{+1} = (c_{+1}, d_{+1}, e_{+1})$ carried into the next DM. The CM problem is

$$\begin{aligned} W_j(\vec{a}) &= \max_{(X^b, h, \vec{a}_{+1})} U(X^b) - h + \beta V_j(\vec{a}_{+1}) \\ \text{subject to } & X^b + \vec{1} \cdot \vec{a}_{+1} = h + \vec{R} \cdot \vec{a} + \tau - \mathbf{1}_{\{d_{+1} > 0\}} \times f - \mathbf{1}_{\{e_{+1} > 0\}} \times \delta, \\ & h \geq 0, c_{+1} \geq 0, d_{+1} \geq 0, \\ & 0 \leq e_{+1} \leq \bar{e}, \end{aligned}$$

where $\vec{1} = (1, 1, 1)$, “ $\cdot$ ” is the inner product, τ is the lump-sum receipt in real terms, f is the fixed cost of holding deposits, and δ is the fixed cost of holding the CBDC.

Assuming an interior solution for h and substituting h from the budget constraint, the CM value function can be written as

$$\begin{aligned} W_j(\vec{a}) &= \vec{R} \cdot \vec{a} + \tau + \max_{X^b} \{U(X^b) - X^b\} \\ &\quad + \max_{\vec{a}_{+1}} \left\{ -\vec{1} \cdot \vec{a}_{+1} + \beta V_j(\vec{a}_{+1}) - \mathbf{1}_{\{d_{+1} > 0\}} \times f - \mathbf{1}_{\{e_{+1} > 0\}} \times \delta \right\} \\ \text{subject to } & c_{+1} \geq 0, d_{+1} \geq 0, 0 \leq e_{+1} \leq \bar{e}. \end{aligned} \tag{1}$$

One standard result of the Lagos-Wright model is $\frac{\partial W_j(\vec{a})}{\partial a} = R^a$ for $a = c, d, e$.

The DM value function $V_j(\vec{a})$ satisfies

$$\begin{aligned} V_j(\vec{a}) &= u(q_j) + W_j(\vec{a} - \vec{p}_j) \\ &= u(q_j) + \vec{R} \cdot (\vec{a}_j - \vec{p}_j) + W_j(\vec{0}), \end{aligned}$$

where $(\vec{p}_j \equiv (p_j^c, p_j^d, p_j^e), q_j)$ are the terms of trade, representing payment in cash, deposits and the CBDC, and the amount of DM good traded in Type j meetings.

Bargaining. For the analytical part, I assume that buyers make a take-it-or-leave-it offer to sellers in the DM, solving

$$\begin{aligned} \max_{q_j, \vec{p}_j} \quad & S^b \\ \text{subject to} \quad & S^s \geq 0, \end{aligned} \tag{2}$$

$$\text{and} \quad 0 \leq \vec{p}_j \leq f_j(\vec{a}), \tag{3}$$

where $S^b = u(q_j) - \vec{R} \cdot \vec{p}_j$, and $S^s = -q_j + \vec{R} \cdot \vec{p}_j$ are buyers' and sellers' DM trade surplus respectively. Specifically, a buyer's trade surplus is her gain from DM consumption minus the value of payment. Constraint (2) is sellers' participation constraint, and (3) is a feasibility constraint that says buyers cannot ask for money from sellers, nor can they offer to transfer more units of money that can be used in Type j meetings than they possess. The function f_j requires that buyers pay with the types of money that are accepted in Type j meetings. For example, if a universal CBDC is introduced, we have $f_1(\vec{a}) = (c, 0, e)$, and $f_2(\vec{a}) = (0, d, e)$.

Solutions to the above problem is

$$q_j = \vec{R} \cdot \vec{p}_j = \begin{cases} \vec{R} \cdot f_j(\vec{a}) & \text{for } \vec{R} \cdot f_j(\vec{a}) < q^*, \\ q^* & \text{for } \vec{R} \cdot f_j(\vec{a}) \geq q^*. \end{cases} \tag{4}$$

Buyers can obtain the socially-efficient quantity q^* if they bring sufficient money to compensate sellers in Type j meetings for the disutility of producing q^* . Otherwise, buyers consume only what their available portfolios allow.

Portfolio choice. Using these results, buyers' portfolio problem in the CM can be written as

$$\begin{aligned} \max_{\vec{a}_{+1}} \quad & \left\{ -\vec{1} \cdot \vec{a}_{+1} + \beta \left[u(q_{j,+1}) + \vec{R} \cdot \vec{a}_{+1} - \vec{R} \cdot \vec{p}_{j,+1} \right] - \mathbf{1}_{\{d_{+1} > 0\}} \times f - \mathbf{1}_{\{e_{+1} > 0\}} \times \delta \right\} \\ \text{subject to} \quad & c_{+1} \geq 0, \quad d_{+1} \geq 0, \quad 0 \leq e_{+1} \leq \bar{e}. \end{aligned}$$

The Lagrangian $\mathcal{L}$ is

$$\mathcal{L} = -\vec{1} \cdot \vec{a}_{+1} + \beta \left[u(q_{j,+1}) + \vec{R} \cdot \vec{a}_{+1} - \vec{R} \cdot \vec{p}_{j,+1} \right] - \mu_1^e (e_{+1} - \bar{e}) + \mu_2^e e_{+1} + \mu^c c_{+1} + \mu^d d_{+1},$$

where $\mu_1^e, \mu_2^e, \mu^c, \mu^d$ are Lagrange multipliers. Define function $\Lambda(L) \equiv \max\{u'(L) - 1, 0\}$, where L is the available liquidity. $\Lambda(L)$ is thus the marginal benefit of liquidity. Recall that buyers know the types of DM meetings they will participate when making portfolio choices, so buyers in Type 1 meetings will not hold deposits and buyers in Type 2 meetings will not accumulate cash. The first-order conditions for cash, deposits and the CBDC are

$$\Lambda(\vec{R} \cdot f_1(\vec{a})) \leq \frac{1 + \mu^c}{\beta R^c} - 1, \quad (5)$$

$$\Lambda(\vec{R} \cdot f_2(\vec{a})) \leq \frac{1 + \mu^d}{\beta R^d} - 1, \quad (6)$$

$$\Lambda(\vec{R} \cdot f_j(\vec{a})) \leq \frac{1 + \mu_1^e + \mu_2^e}{\beta R^e} - 1, \quad (7)$$

which hold with equality if and only if $a > 0$ for $a = c, d, e$. The complementary slackness conditions for $0 \leq e_{+1} \leq \bar{e}$ are

$$\mu_1^e, \mu_2^e \geq 0, \quad (8)$$

$$\mu_1^e(e_{+1} - \bar{e}) = 0, \quad (9)$$

$$\mu_2^e e_{+1} = 0. \quad (10)$$

The complementary slackness conditions for $c_{+1} \geq 0$ and $d_{+1} \geq 0$ are

$$\mu^c, \mu^d \geq 0, \quad (11)$$

$$\mu^c c_{+1} = 0, \quad (12)$$

$$\mu^d d_{+1} = 0. \quad (13)$$

Equations (5)-(13) thus characterize the demand for each type of money in the period- t CM.

2.3 Asset supply

Currency. I focus on stationary equilibria where the total supply of central bank monies M —including both paper and digital currencies—grows at a constant rate π , such that $\frac{M}{M_{-1}} = \frac{\phi_{-1}}{\phi} = \pi > \beta$. Here it is implicitly assumed that the central bank controls only the total supply of cash and CBDC, but not the composition, and households can exchange one for the other at a fixed exchange rate of one. Therefore, the central bank's budget constraint is

$$\phi(C + E) = \phi(C_{-1} + (1 + i)E_{-1}) + \tau - (\lambda_1 \lambda_1^e + \lambda_2 \lambda_2^e) \delta_2,$$

where C and E are the amounts of cash and CBDC outstanding, τ is the real transfer to households, and $(\lambda_1\lambda_1^e + \lambda_2\lambda_2^e)\delta_2$ is the service charge (subsidies) paid by (to) CBDC users, where λ_1^e and λ_2^e are fractions of Type 1 and Type 2 meetings where the CBDC is used.

Deposits. As in Keister and Sanches (2022), the banking sector is perfectly competitive with limited commitment. Given a market deposit rate R^d , a banker with project return $\gamma \in [0, \bar{\gamma}]$ is willing to issue deposits and invest only if

$$\frac{1}{1-\mu}R^d \leq \gamma + \frac{\mu}{(1-\mu)\pi},$$

where the right-hand side is the banker's investment income. For $\frac{1}{1-\mu}$ units of deposits issued, one unit is invested in the project which generates γ units of the CM good next period. This explains the first term. The remaining $\frac{\mu}{1-\mu}$ units of issued deposits is invested in reserves with a rate of return of $\frac{1}{\pi}$, which is the second term. In addition, as bankers have limited commitment, they cannot promise to repay depositors more than the value of their pledgeable collateral

$$\frac{1}{1-\mu}R^d \leq \epsilon\gamma + \frac{\mu}{(1-\mu)\pi}, \quad (14)$$

where I assume bankers' asset holdings of central bank reserves is fully pledgeable. Let $\hat{\gamma}$ denote the productivity of the marginal bank who satisfies the pledgeability constraint (14) at equality. Thus I have

$$R^d = (1-\mu)\epsilon\hat{\gamma} + \frac{\mu}{\pi}. \quad (15)$$

The aggregate supply of deposits is the total amount of deposits issued by bankers with productivity at least $\hat{\gamma}$, which is $\frac{\eta}{1-\mu}(\bar{\gamma} - \hat{\gamma})$. Note that bankers who are able to issue deposits will issue $\frac{1}{1-\mu}$ units of deposits so that they can invest exactly one unit of CM good into their projects after satisfying the reserve requirement. Equation (15) suggests that given the equilibrium deposit rate R^d , a smaller ϵ leads to a higher $\hat{\gamma}$, which means that fewer bankers are able to issue deposits and thereby invest. Meanwhile, given ϵ , the higher the deposit rate R^d is, the higher the cutoff productivity $\hat{\gamma}$ is and thereby the lower the level of investment is.

2.4 Market clearing

Since cash is used only in Type 1 meetings and deposits are only used in Type 2 meetings, their market clearing conditions in real terms are

$$\phi C = \lambda_1 \lambda_1^c c + \mu \lambda_2 (\lambda_2^d d + \lambda_2^{mix} \tilde{d}), \quad (16)$$

$$\frac{\eta}{1 - \mu} (\bar{\gamma} - \hat{\gamma}) = \lambda_2 (\lambda_2^d d + \lambda_2^{mix} \tilde{d}), \quad (17)$$

where λ_1^c is the fraction of Type 1 meetings where cash is used, λ_2^d is the fraction of Type 2 meetings where *only* deposits are used, and λ_2^{mix} where the mixture of deposits and the CBDC is used. d and $\tilde{d}$ are households' demand for deposits when they hold only deposits, and hold both deposits and the CBDC. I distinguish between the two, as they can coexist in equilibrium. Reserves put in the central bank by private banks are held in the form of cash, hence the second term in the demand side for cash. The market clearing condition for the CBDC is

$$\phi E = (\lambda_1 \lambda_1^e + \lambda_2 \lambda_2^e) e. \quad (18)$$

Definition 1. *A stationary equilibrium is a list of portfolios accumulated by buyers $\{\vec{a}_j\}_{j=1,2}$, terms of trade $\{\vec{p}_j, q_j\}_{j=1,2}$, fractions of buyers holding different portfolios $\{\lambda_1^c, \lambda_1^e, \lambda_2^d, \lambda_2^e, \lambda_2^{mix}\} \in [0, 1]$ where $\lambda_2^d + \lambda_2^e = 1$, the deposit rate R^d and the cutoff bank's productivity $\hat{\gamma}$ that satisfy equations (4)-(13), (15)-(18) given policy parameters $\{\pi, i, \delta, \bar{e}\}$.*

In what follows, I will characterize the equilibrium in the benchmark economy where there is no CBDC in Section 3. Then, I will introduce a CBDC into the economy and derive households' portfolio choices for both cases when the CBDC holding limit is not binding in Section 4.1 and is binding in Section 4.2.

3 Equilibrium With No Digital Currency

Without a CBDC, buyers can only use cash in Type 1 meetings and only use deposits in Type 2 meetings, i.e., $f_1(\vec{a}) = (c, 0, 0)$, $f_2(\vec{a}) = (0, d, 0)$ in feasibility constraint (3). Therefore, the corresponding first-order conditions for cash and deposits (5)-(6) will hold with equality:

$$\Lambda(R^c c(R^c)) = \frac{1}{\beta R^c} - 1, \quad (19)$$

$$\Lambda(R_0^d d(R_0^d)) = \frac{1}{\beta R_0^d} - 1, \quad (20)$$

where the subscript 0 in R_0^d suggests the benchmark equilibrium deposit rate when there is no CBDC.

Using equation (15) and the market clearing conditions (16)-(17), conditions (19)-(20) can be written as functions of $c(R^c)$ and R_0^d respectively:

$$\Lambda\left(\frac{c(R^c)}{\pi}\right) = \frac{\pi}{\beta} - 1, \quad (21)$$

$$\Lambda\left(R_0^d \frac{\eta(\bar{\gamma} - \frac{R_0^d - \frac{\mu}{(1-\mu)\epsilon}}{\lambda_2})}{\lambda_2}\right) = \frac{1}{\beta R_0^d} - 1. \quad (22)$$

To guarantee the existence and uniqueness of equilibrium, I make the following assumption that is maintained throughout the rest of the paper:

Assumption 1. *Preferences are such that:*

$$(i) \ d(R^d) \text{ is strictly increasing in } R^d \quad \text{and} \quad (ii) \ \lim_{R^d \rightarrow 0} \lambda_2 \frac{\lambda^{-1} \left(\frac{1}{\beta R^d} - 1 \right)}{R^d} < \eta(\bar{\gamma} - \hat{\gamma}(R^d)).$$

Assumption 1 ensures that the supply of and demand for deposits intersect at a unique deposit rate, which is well-defined. The following proposition defines the equilibrium without CBDC.

Proposition 1. *In the benchmark economy without CBDC, there exists a unique equilibrium consisting of portfolios $\{\vec{a}_1 = (c(R^c), 0, 0), \vec{a}_2 = (0, d(R_0^d), 0)\}$, terms of trade $\{\vec{p}_j, q_j\}_{j=1,2}$, and a deposit rate R_0^d satisfying equations (4) and (19)-(22) given parameter π .*

When $\epsilon < 1$, the benchmark equilibrium can feature underinvestment— $\hat{\gamma} > \frac{1}{\beta}$ —and introducing a CBDC that competes with deposits and raises the deposit rate consequently can further crowd out investment, although the efficiency of DM trade improves. This trade-off between CM investment and DM trade is at the core of the analysis in the following sections.

4 Equilibrium With A Digital Currency

In this section, I look at the equilibrium when a CBDC is introduced and it competes with both cash and deposits in general. I assume that the CBDCs used in Type 1 and Type 2 meetings have the same design: the interest rate R^e , the holding limit $\bar{e}$, and the adoption cost δ . I will derive buyers' optimal portfolio choices in the CM given different designs of the CBDC in subsequent sections. In Section 4.1, the rate of return on the CBDC is sufficiently

low or the holding limit on the CBDC is sufficiently high such that the amount of the CBDC that households carry, if they use it, does not exceed the limit. And the opposite in Section 4.2.

In Type 1 meetings, buyers choose the portfolio $\vec{a}_1 = (c, 0, e)$ that maximizes the following CM value function obtained by rearranging and collecting terms from expression (1)

$$\begin{aligned} W_1(\vec{a}_1) = & (\beta R^c - 1)c + (\beta R^e - 1)e \\ & + \beta (u(q_1^c + q_1^e) - q_1^c - q_1^e) - \mathbf{1}_{\{e > 0\}} \times \delta + \Omega_1, \end{aligned} \quad (23)$$

where $\Omega_1 = \vec{R} \cdot \vec{a} + \tau + \max_{X^b} \{U(X^b) - X^b\} + \beta W_{1,+1}(\vec{0})$ is independent of buyers' portfolio choices, q_1^c is the amount of DM goods paid by cash and q_1^e by the CBDC.

Similarly, in Type 2 meetings, buyers choose the portfolio $\vec{a}_2 = (0, d, e)$ that maximizes the following CM value function.

$$\begin{aligned} W_2(\vec{a}_2) = & (\beta R^d - 1)d + (\beta R^e - 1)e \\ & + \beta (u(q_2^d + q_2^e) - q_2^d - q_2^e) - \mathbf{1}_{\{d > 0\}} \times f - \mathbf{1}_{\{e > 0\}} \times \delta + \Omega_2, \end{aligned} \quad (24)$$

where $\Omega_2 = \vec{R} \cdot \vec{a} + \tau + \max_{X^b} \{U(X^b) - X^b\} + \beta W_{2,+1}(\vec{0})$, q_2^d is the amount of DM goods consumed using deposits and q_2^e using the CBDC, and f is the deposit fixed cost.

4.1 Non-binding CBDC holding limit

In this section, the central bank sets $(R^e, \bar{e})$ such that $e(R^e) \leq \bar{e}$, where $e(R^e)$ is obtained from the first-order condition (7) holding with equality: $\Lambda(R^e e(R^e)) = \frac{1}{\beta R^e} - 1$.

We start with Type 1 meetings. The following lemma states that it will never be optimal for buyers to accumulate both cash and the CBDC in a single portfolio since the only feasible case is when the two types of monies have the same rates of return— $R^c = R^e$ —and when this is the case, buyers will optimally choose to bring only cash given a strictly positive fixed cost of using the CBDC. Proofs of all propositions and lemmata are contained in the Appendix.

Lemma 1. *When $e(R^e) \leq \bar{e}$, in Type 1 meetings, given $\delta > 0$, either $\vec{a}_1 = (c(R^e), 0, 0)$, or $\vec{a}_1 = (0, 0, e(R^e))$.*

Therefore, buyers choose between the cash-only portfolio and the CBDC-only portfolio, and pick whichever has a higher CM value. Denote W_1^c and W_1^e as the optimal CM values

for holding only cash and only the CBDC, respectively. From (23), we have

$$W_1^c = (\beta R^c - 1)c(R^c) + \beta (u(q_1^c) - q_1^c) + \Omega_1, \quad (25)$$

$$W_1^e = (\beta R^e - 1)e(R^e) + \beta (u(q_1^e) - q_1^e) - \delta + \Omega_1. \quad (26)$$

With (25)-(26), I obtain a threshold of R^e , which is a function of policy parameters R^c and δ , and I denote it as $R^e(R^c, \delta)$. $R^e(R^c, \delta)$ is the rate of return on the CBDC that makes it as desirable as cash to buyers, which strictly increases in R^c and δ . Given a strictly positive fixed cost of using the CBDC, the CBDC will have to pay a positive net nominal interest rate to incentivize buyers to switch from using cash to its digital counterpart. So were the CBDC to be adopted, it would be such that $e(R^e) > c(R^c)$ and $q_1^e > q_1^c$. Hence, when deciding whether to adopt the CBDC, households face a tradeoff between more efficient DM exchange and a higher fixed cost. The following proposition dictates buyers' portfolio choices in Type 1 meetings when R^e takes different ranges of value.

Proposition 2. *When $e(R^e) \leq \bar{e}$, in Type 1 meetings, given $\delta > 0$, there exists a cutoff $R^e(R^c, \delta) > R^c$ such that*

- (i) *when $R^e < R^e(R^c, \delta)$, $\vec{a}_1 = (c(R^c), 0, 0)$, and $q_1 = q_1^c$;*
- (ii) *when $R^e > R^e(R^c, \delta)$, $\vec{a}_1 = (0, 0, e(R^e))$, $e(R^e) > c(R^c)$, and $q_1 = q_1^e > q_1^c$;*
- (iii) *when $R^e = R^e(R^c, \delta)$, a fraction $\lambda_1^c \in [0, 1]$ of households choose $\vec{a}_1 = (c(R^c), 0, 0)$ with DM consumption q_1^c , while the remaining fraction $1 - \lambda_1^c$ choose $\vec{a}_1 = (0, 0, e(R^e))$ with q_1^e , where $e(R^e) > c(R^c)$ and $q_1^e > q_1^c$.*

Proof. See the Appendix.

Now, I turn to Type 2 meetings. It again will not be optimal for buyers to hold a mixed portfolio of deposits and the CBDC, as when they have the same rates of return, buyers will choose the one with a lower fixed cost. Lemma 2 formalizes this idea.

Lemma 2. *When $e(R^e) \leq \bar{e}$, in Type 2 meetings, given $f, \delta > 0$, either $\vec{a}_2 = (0, d(R^d), 0)$, or $\vec{a}_2 = (0, 0, e(R^e))$.*

Thus, buyers will choose between the deposit-only and CBDC-only portfolios whichever has a higher CM value. Denote $W_2^d(R^d)$ and W_2^e as the optimal CM values for holding only deposits at deposit rate R^d and holding only the CBDC, respectively.

$$W_2^d(R^d) = (\beta R^d - 1)d(R^d) + \beta (u(q_2^d(R^d)) - q_2^d(R^d)) - f + \Omega_2, \quad (27)$$

$$W_2^e = (\beta R^e - 1)e(R^e) + \beta (u(q_2^e) - q_2^e) - \delta + \Omega_2. \quad (28)$$

If the CBDC is not adopted in Type 2 meetings, then we have $R^d = R_0^d$, the equilibrium deposit rate in the benchmark economy without a CBDC. A key difference here with Type 1 meetings is that when the CBDC offers a higher CM value than deposits with deposit rate R_0^d , the CBDC would not drive out deposits completely in Type 2 meetings. Instead, the deposit rate will increase until, in equilibrium, the CM value of holding deposits matches the CM value of holding the CBDC, given the following assumptions.

Assumption 2.

$$(i) \quad W_2^e \leq W_2^d(R^d = \frac{1}{\beta}) \quad \text{and} \quad (ii) \quad \bar{\gamma} > \hat{\gamma}(R^d).$$

Assumption 2 ensures that deposits will always be used in equilibrium. Therefore, with (27) evaluated at R_0^d and (28), I obtain a threshold of R^e , which depends on R_0^d and δ , and I denote it as $R^e(R_0^d, \delta)$. $R^e(R_0^d, \delta)$ is the CBDC rate of return that makes it as desirable as deposits in the benchmark economy to buyers, which strictly increases in R_0^d and δ . Proposition 3 specifies households portfolio choices in Type 2 meetings given different values of R^e and Proposition 4 gives some comparative statics regarding R^e .

Proposition 3. *When $e(R^e) \leq \bar{e}$, in Type 2 meetings, there exists a cutoff $R^e(R_0^d, \delta)$ such that*

- (i) *when $R^e \leq R^e(R_0^d, \delta)$, $R^d = R_0^d$, $\vec{a}_2 = (0, d(R_0^d), 0)$, and $q_2 = q_2^d(R_0^d)$;*
- (ii) *when $R^e(R_0^d, \delta) < R^e \leq \frac{1}{\beta}$, a fraction $\lambda_2^d \in (0, 1)$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$ with DM consumption $q_2^d(R^d)$, while the remaining fraction $1 - \lambda_2^d$ choose $\vec{a}_2 = (0, 0, e(R^e))$ with q_2^e . Here, R^d satisfies $W_2^d(R^d) = W_2^e$ and $R^d > R_0^d$.*

Proof. See the Appendix.

Proposition 4. *When $e(R^e) \leq \bar{e}$, if the CBDC is adopted in at least one type of meetings such that*

- (i) *$R^e > \max \{R^e(R^c, \delta), R^e(R_0^d, \delta)\}$, then as R^e increases, $q_1, q_2^e, R^d, q_2^d, \lambda_2^e$ also increase, while λ_2^d decreases;*
- (ii) *$R^e(R^c, \delta) < R^e < R^e(R_0^d, \delta)$, then as R^e increases, q_1 also increases, while R^d, q_2^d, λ_2^d remain unchanged;*
- (iii) *$R^e(R_0^d, \delta) < R^e < R^e(R^c, \delta)$, then as R^e increases, $q_2^e, R^d, q_2^d, \lambda_2^e$ also increase, while λ_2^d decreases and q_1 remains unchanged.*

Proof. See the Appendix.

4.2 Binding CBDC holding limit

In this section, the central bank sets $(R^e, \bar{e})$ such that $\bar{e} < e(R^e)$. As we have a binding CBDC limit, buyers in either type of meetings may choose to hold both means of payment available in a single portfolio. The idea is that when the rate of return on the CBDC is high and the fixed cost of holding the CBDC is small, buyers will start accumulating the CBDC up to the holding limit $\bar{e}$, and when the limit is low, they will continue topping up with cash in Type 1 meetings, and with deposits in Type 2 meetings if the fixed cost for using deposits is not too large, to have more DM trade from this extra liquidity.

Type 1 meetings. In Type 1 meetings, there are three possible types of portfolios that buyers may choose. The following lemma outlines them.

Lemma 3. *When $\bar{e} < e(R^e)$, in Type 1 meetings, buyers may choose one of the following portfolios*

- (i) $\vec{a}_1 = (c(R^c), 0, 0)$;
- (ii) $\vec{a}_1 = (0, 0, \bar{e})$;
- (iii) $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, where $\tilde{c} = c(R^c) - \frac{R^e \bar{e}}{R^c}$. The mixed portfolio is feasible only if $R^e > R^c$ and $R^e \bar{e} < R^c c(R^c)$.

Buyers may hold only cash, hold the CBDC up to the holding limit, or hold the CBDC up to the limit and supplement it with cash. Regarding the two feasibility conditions for the third portfolio $(\tilde{c}, 0, \bar{e})$, where cash coexists with the CBDC at the intensive margin, the former one— $R^e > R^c$ —implies that buyers prefer the CBDC at the margin as it has a higher rate of return, but the amount of the CBDC they are allowed to hold is sufficiently low such that the marginal gain of DM consumption exceeds the marginal cost of holding cash, as implied by the latter condition— $R^e \bar{e} < R^c c(R^c)$. Buyers will choose to hold the portfolio that gives them the highest value when they make their portfolio decisions in the CM. The CM values for these three types of portfolios are:

$$\begin{aligned} W_1^c &= (\beta R^c - 1)c(R^c) + \beta(u(q_1^c) - q_1^c) + \Omega_1, \\ W_1^{\bar{e}} &= (\beta R^e - 1)\bar{e} + \beta(u(q_1^{\bar{e}}) - q_1^{\bar{e}}) - \delta + \Omega_1, \\ W_1^{mix} &= (\beta R^e - 1)\bar{e} + (\beta R^c - 1)\tilde{c} + \beta(u(q_1^{mix}) - q_1^{mix}) - \delta + \Omega_1, \end{aligned}$$

where the superscript $\bar{e}$ indicates that buyers hold $\bar{e}$ units of the CBDC, $q_1^{\bar{e}}$ is the amount of DM goods purchased using $\bar{e}$ units of the CBDC and q_1^{mix} using both cash and the CBDC, and $\tilde{c}$ is the amount of cash held in the mixed portfolio.

Note two things here. One is that when the mixed portfolio is feasible, we have $W_1^{mix} > W_1^{\bar{e}}$, as there is no fixed cost for using cash and the amount of liquidity provided by CBDC is sufficiently low to make topping up with cash desirable. In other words, when cash can coexist with the CBDC at the intensive margin, buyers will never choose the CBDC-only portfolio in Type 1 meetings. Lemma 4 summarizes this result. The other is that the amount of liquidity, and thereby the amount of DM goods traded, is the same for the cash-only portfolio and the mixed portfolio— $R^e \bar{e} + R^c \tilde{c} = R^c c(R^c)$. However, by accumulating the CBDC to the limit and topping up with cash in the mixed portfolio, buyers, on one hand, hold fewer units of money in real terms and thereby save in their efforts to acquire money— $\tilde{c} + \bar{e} < c(R^c)$ —since the CBDC pays a higher interest rate than cash; on the other hand, they incur a fixed cost for using the CBDC. Lemma 5 records these results. Whether it is optimal to do so depends on which one is greater: savings in labor to acquire money or the CBDC fixed cost. If $i\bar{e} > (<) \delta$, then the mixed portfolio is better (worse) than the cash-only portfolio.

Lemma 4. *When $\bar{e} < e(R^e)$, in Type 1 meetings, if $R^e > R^c$ and $R^e \bar{e} < R^c c(R^c)$, then buyers will never choose $\vec{a}_1 = (0, 0, \bar{e})$.*

Proof. See the Appendix.

Lemma 5. *When $\bar{e} < e(R^e)$, in Type 1 meetings, if $R^e > R^c$ and $R^e \bar{e} < R^c c(R^c)$, then $R^e \bar{e} + R^c \tilde{c} = R^c c(R^c)$, $q_1^{mix} = q_1^c$, $\tilde{c} + \bar{e} < c(R^c)$, and $W_1^{mix} = W_1^c + i\bar{e} - \delta$.*

Proof. See the Appendix.

Next, I will derive buyers' optimal portfolio choices in Type 1 meetings. There are three cases. First, when the CBDC is not paying a higher interest rate than cash, the CBDC will not be adopted given its fixed cost. Second, if the CBDC pays a positive net nominal interest rate but its holding limit is too high for the mixed portfolio to be feasible, buyers will choose between the cash-only portfolio and the CBDC-only portfolio. Third, if the CBDC pays a positive net nominal interest rate and its holding limit is sufficiently low to make the mixed portfolio feasible, buyers will choose between the cash-only portfolio and the mixed portfolio, and buyers will opt for the mixed portfolio if the limit is sufficiently large such that it is worth paying the CBDC fixed cost to save on the cost of acquiring money. The following lemma defines some important cutoffs for the CBDC holding limit and shows how the cutoffs position relative to each other.

Lemma 6. *When $\bar{e} < e(R^e)$, I obtain three cutoffs— $\bar{e}_1, \bar{e}_2, \bar{e}_3$ —such that:*

$$(i) \ \bar{e}_1 \equiv \frac{\delta}{i}, \text{ and } W_1^c = W_1^{mix}|_{\bar{e}=\bar{e}_1};$$

(ii) $\bar{e}_2 \equiv \bar{e}(R^c, R^e, \delta)$, and $W_1^c = W_1^{\bar{e}=\bar{e}_2}$;

(iii) $\bar{e}_3 \equiv \frac{R^c c(R^c)}{R^e}$, and $\tilde{c}(\bar{e}_3) = 0$.

And when $R^e > R^c$, there exists $\hat{\delta} > 0$ such that:

(i) if $0 < \delta < \hat{\delta}$, then $\bar{e}_1 < \bar{e}_2 < \bar{e}_3$;

(ii) if $\delta \geq \hat{\delta}$, then $\bar{e}_3 \leq \min\{\bar{e}_1, \bar{e}_2\}$, where $\hat{\delta} = (1 - \frac{R^c}{R^e})c(R^c)$.

Proof. See the Appendix.

$\bar{e}_1$ is the cutoff that the holding limit must exceed for the mixed portfolio to outperform the cash-only portfolio, $\bar{e}_2$ denotes the threshold that makes the CBDC-only portfolio as desirable as the cash-only portfolio, and $\bar{e}_3$ specifies the upper bound on the limit for the mixed portfolio to be feasible. If the limit surpasses $\bar{e}_3$, the amount of cash held in the mixed portfolio becomes negative, implying that households would take a short position in cash, which is not allowed.

The following proposition consolidates these results and presents buyers' optimal portfolio choices in Type 1 meetings under different designs of the CBDC.

Proposition 5. *When $\bar{e} < e(R^e)$, in Type 1 meetings,*

1. $\vec{a}_1 = (c(R^c), 0, 0)$, $q_1 = q_1^c$ when:

(i) $R^e \leq R^c$, or

(ii) $R^e > R^c$, $0 < \delta < \hat{\delta}$, and $\bar{e} < \bar{e}_1$, or

(iii) $R^e > R^c$, $\delta \geq \hat{\delta}$, and $\bar{e} < \bar{e}_2$;

2. $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, $q_1 = q_1^{mix}$ when $R^e > R^c$, $0 < \delta < \hat{\delta}$, and $\bar{e}_1 < \bar{e} < \bar{e}_3$;

3. $\vec{a}_1 = (0, 0, \bar{e})$, $q_1 = q_1^{\bar{e}}$ when $R^e > R^c$, and $\bar{e} > \max\{\bar{e}_2, \bar{e}_3\}$.

Proof. See the Appendix.

Figures 2 and 3 summarize how buyers' optimal portfolio choices in Type 1 meetings change as the CBDC holding limit varies, for the two cases: (i) $R^e > R^c$, $0 < \delta < \hat{\delta}$; and (ii) $R^e > R^c$, $\delta \geq \hat{\delta}$. In Figure 2, $0 < \delta < \hat{\delta}$, so buyers switch from holding only cash, to holding a mixture of cash and the CBDC, and lastly to holding only the CBDC, as the limit relaxes. In Figure 3, $\delta \geq \hat{\delta}$, and therefore $\bar{e}_1 \geq \bar{e}_3$, which implies that the range of holding limits making the mixed portfolio optimal lies outside its feasible set. In other words, whenever the mixed portfolio is feasible, it is not optimal. Therefore, buyers switch from holding only cash to holding only the CBDC, as the limit increases.

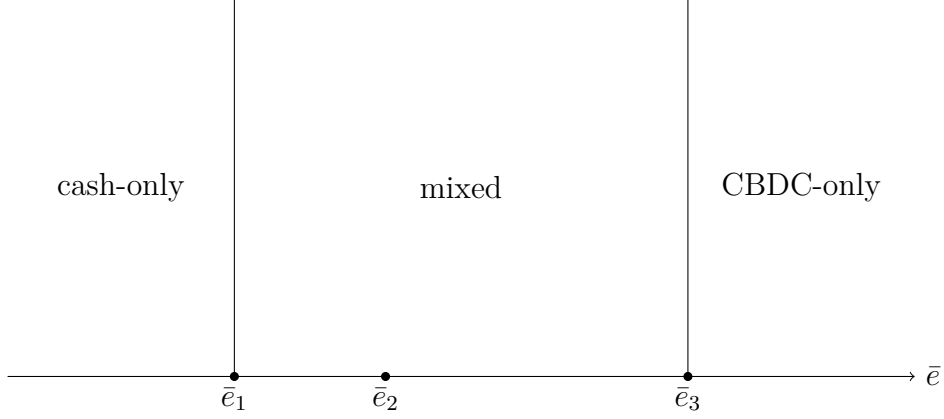


Figure 2: Optimal portfolio in Type 1 meetings when $R^e > R^c$ and $0 < \delta < \hat{\delta}$.

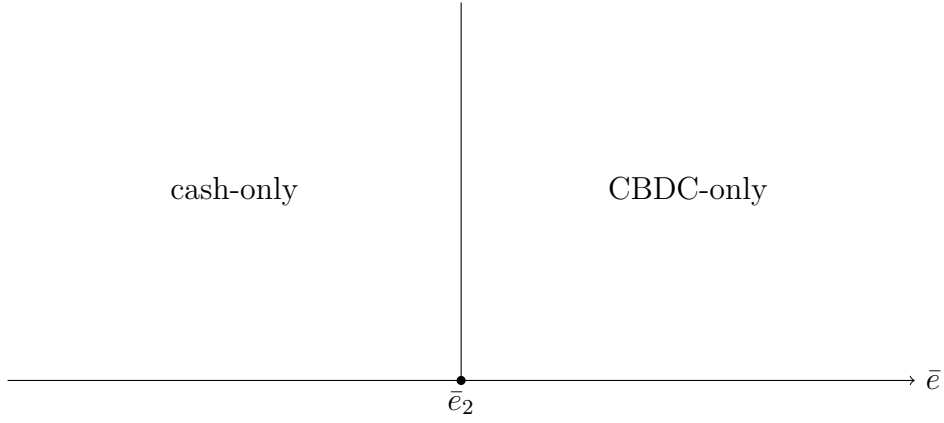


Figure 3: Optimal portfolio in Type 1 meetings when $R^e > R^c$ and $\delta \geq \hat{\delta}$.

Type 2 meetings. In Type 2 meetings, there are also three possible types of portfolios that buyers can hold. Individual buyers may hold only deposits, hold only the CBDC up to the holding limit, or hold a mixture of the two types of money. Let $\tilde{d}$ denote the amount of deposits in the mixed portfolio. The optimal CM values of the three types of portfolios are:

$$\begin{aligned}
 W_2^d(R^d) &= (\beta R^d - 1)d(R^d) + \beta (u(q_2^d(R^d)) - q_2^d(R^d)) - f + \Omega_2, \\
 W_2^{\bar{e}} &= (\beta R^e - 1)\bar{e} + \beta (u(q_2^{\bar{e}}) - q_2^{\bar{e}}) - \delta + \Omega_2, \\
 W_2^{mix}(R^d) &= (\beta R^e - 1)\bar{e} + (\beta R^d - 1)\tilde{d} + \beta (u(q_2^{mix}(R^d)) - q_2^{mix}(R^d)) - \delta - f + \Omega_2.
 \end{aligned}$$

The mixed portfolio $(0, \tilde{d}, \bar{e})$, is feasible only if

$$(i) \quad R^e > R^d \quad \text{and} \quad (ii) \quad R^e \bar{e} < R^d d(R^d). \quad (29)$$

When the mixed portfolio is feasible, compared with the deposit-only portfolio at the same deposit rate, the DM consumption is the same— $q_2^{mix} = q_2^d(R^d)$. However, the mixed portfolio saves on labor to acquire money as $\tilde{d} + \bar{e} < d(R^d)$, by paying the fixed cost for using the CBDC. Lemma 7 documents these results.

Lemma 7. *When $\bar{e} < e(R^e)$, in Type 2 meetings, if $R^e > R^d$ and $R^e \bar{e} < R^d d(R^d)$, then $R^e \bar{e} + R^d \tilde{d} = R^d d(R^d)$, $q_2^{mix} = q_2^d(R^d)$, $\tilde{d} + \bar{e} < d(R^d)$, and $W_2^{mix}(R^d) = W_2^d(R^d) + (\frac{R^e}{R^d} - 1)\bar{e} - \delta$.*

Proof. See the Appendix.

To guarantee the monotonicity of equilibrium portfolios in the holding limit, I make the following assumptions

Assumption 3.

$$(i) \quad \frac{\partial (W_2^{\bar{e}} - W_2^{mix}(R^d))}{\partial \bar{e}} > 0 \quad \text{and} \quad (ii) \quad \frac{\partial R^d \tilde{d}}{\partial \bar{e}} < 0.^9$$

In general, there are two cases for households' optimal portfolio choices in Type 2 meetings. If the CBDC is not paying a higher interest rate than R_0^d , households either hold only deposits or hold only the CBDC. In this case, when the limit is set sufficiently low, the CBDC will not be adopted in Type 2 meetings. When the limit is set sufficiently large, it creates competitive pressure on banks to raise the deposit rate until households become indifferent between holding only deposits or only the CBDC. At this point, a nonnegative fraction of households switch to holding the CBDC from deposits as fewer banks can afford to issue deposits to finance their investment, and we have the extensive margin coexistence of deposits and the CBDC. If, instead, the CBDC offers a higher interest payment, households can hold the mixed portfolio. To facilitate analysis, Lemma 8 defines four cutoffs for $\bar{e}$ and shows their relative positions for the case when $R^e > R_0^d$.

Lemma 8. *When $\bar{e} < e(R^e)$, I obtain four cutoffs— $\bar{e}'_1, \bar{e}'_2, \bar{e}'_3, \bar{e}'_4$ —such that:*

- (i) $\bar{e}'_1 \equiv \frac{\delta}{\frac{R^e}{R_0^d} - 1}$, and $W_2^d(R_0^d) = W_2^{mix}(R_0^d)|_{\bar{e}=\bar{e}'_1}$;
- (ii) $\bar{e}'_2 \equiv \bar{e}(R_0^d, R^e, \delta, f)$, and $W_2^d(R_0^d) = W_2^{\bar{e}=\bar{e}'_2}$;
- (iii) $\bar{e}'_3 \equiv \bar{e}(R^e, \delta, f)$, and $W_2^{mix}(R^d)|_{\bar{e}=\bar{e}'_3} = W_2^{\bar{e}=\bar{e}'_3}$;
- (iv) $\bar{e}'_4 \equiv \bar{e}(R^e, \delta)$, and $\tilde{d}(\bar{e}'_4) = 0$.

⁹Under CRRA form for $u(q)$, a necessary condition for (i) to hold is $\beta(R^e)^{1-\sigma} \geq \bar{e}^\sigma$, and when $\frac{\delta}{(\delta+\bar{e})^2} < \beta$, (ii) holds for $\delta \geq 1$.

And when $R^e > R_0^d$ and $W_2^d(R_0^d, f = 0) > W_2^{\bar{e}=\bar{e}'_1}$, there exist $\hat{f} > 0$ such that:

(i) if $f > \hat{f}$, then $\bar{e}'_2 < \bar{e}'_1$;

(ii) if $0 < f < \hat{f}$, then $\bar{e}'_1 < \bar{e}'_2 < \bar{e}'_3 < \bar{e}'_4$, where $W_2^d(R_0^d, \hat{f}) = W_2^{\bar{e}=\bar{e}'_1}$.

Proof. See the Appendix.

When the equilibrium deposit rate is R_0^d , buyers are indifferent between holding the mixture and holding only deposits at $\bar{e}'_1$, while they are indifferent between holding only the CBDC and holding only deposits at $\bar{e}'_2$. Beyond $\bar{e}'_3$, buyers no longer find it optimal to top up with deposits after already accumulating the CBDC to the holding limit, as the extra amount of liquidity deposits provide can no longer compensate for its fixed cost, whereas beyond $\bar{e}'_4$, the mixed portfolio collapses to containing only the CBDC.

Proposition 6 records these results and shows buyers' optimal portfolio choices in Type 2 meetings under different designs of the CBDC.

Proposition 6. *When $\bar{e} < e(R^e)$, in Type 2 meetings, the CBDC is adopted in three distinct ways:*

1. $\vec{a}_2 = (0, d(R_0^d), 0)$, $q_2 = q_2^d(R_0^d)$, and $R^d = R_0^d$ when:

(i) $R^e \leq R_0^d$ and $\bar{e} \leq \bar{e}'_2$, or

(ii) $R^e > R_0^d$ and $\bar{e} \leq \min\{\bar{e}'_1, \bar{e}'_2\}$;

2. a fraction $\lambda_2^d \in (0, 1)$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$ with DM consumption $q_2^d(R^d)$, while the remaining fraction $1 - \lambda_2^d$ choose $\vec{a}_2 = (0, \tilde{d}, \bar{e})$ with $q_2^{mix} = q_2^d(R^d) > q_2^d(R_0^d)$, where $R^d = \frac{R^e \bar{e}}{\delta + \bar{e}} > R_0^d$, when $R^e > R_0^d$, $0 < f < \hat{f}$, and $\bar{e}'_1 < \bar{e} < \bar{e}'_3$;

3. a fraction $\lambda_2^d \in (0, 1)$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$ with DM consumption $q_2^d(R^d)$, while the remaining fraction $1 - \lambda_2^d$ choose $\vec{a}_2 = (0, 0, e(R^e))$ with q_2^e , where R^d is such that $W_2^d(R^d) = W_2^e$ and $R^d > R_0^d$ when:

(i) $R^e \leq R_0^d$ and $\bar{e} > \bar{e}'_2$, or

(ii) $R^e > R_0^d$ and $\bar{e} > \max\{\bar{e}'_2, \bar{e}'_3\}$.

Proof. See the Appendix.

Figures 4 and 5 summarize how buyers' optimal portfolio choices in Type 2 meetings change as the CBDC holding limit varies, for the two cases: (i) $R^e > R_0^d$, $0 < f < \hat{f}$; and (ii) $R^e > R_0^d$, $f \geq \hat{f}$. In Figure 4, where $0 < f < \hat{f}$, buyers initially do not adopt the CBDC.

As the holding limit relaxes, they first become indifferent between holding only deposits and holding the mixed portfolio, and eventually become indifferent between holding only deposits and holding only the CBDC as it is no longer worth paying the fixed cost of deposits and holding the mixture. In Figure 5, where $f \geq \hat{f}$, we have $\bar{e}'_1 \geq \bar{e}'_2$, suggesting that the fixed cost for using deposits is too large to make it optimal for households to hold both deposits and the CBDC regardless of the value of the limit. Consequently, buyers optimally choose between the deposit-only portfolio and the CBDC-only portfolio.

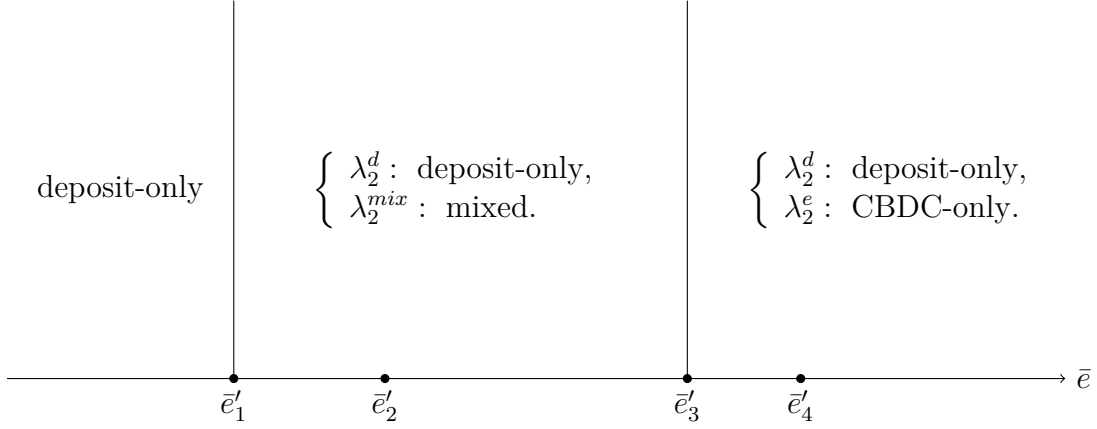


Figure 4: Optimal portfolio in Type 2 meetings when $R^e > R_0^d$ and $0 < f < \hat{f}$.

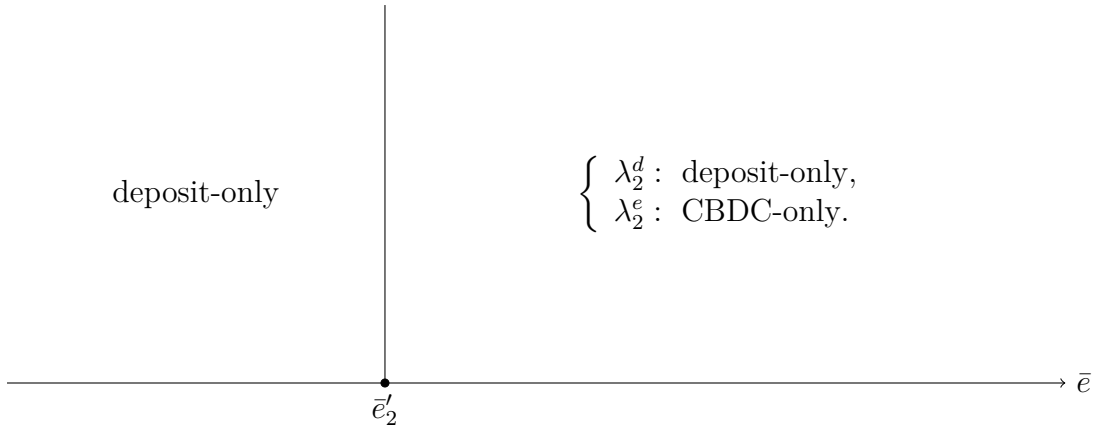


Figure 5: Optimal portfolio in Type 2 meetings when $R^e > R_0^d$ and $f \geq \hat{f}$.

If the CBDC is adopted in Type 2 meetings, a higher CBDC holding limit until it stops binding always results in a higher deposit rate, which makes fewer banks able to issue deposits and invest. However, the rate at which a higher CBDC holding limit raises the deposit rate—thereby crowding out deposits and investment—differs depending on how the CBDC coexists with deposits in equilibrium. In Section 5.3, I conduct a quantitative analysis

and investigate the difference.

Proposition 7 integrates the results for both Type 1 and Type 2 meetings.

Proposition 7. *When $\bar{e} < e(R^e)$, if $R^e > \max\{R^c, R_0^d\}$, $R_0^d > R^c > \hat{R}^c$, $W_2^d(R_0^d, f = 0) > W_2^{\bar{e}=\bar{e}'_1}$, $0 < \delta < \hat{\delta}$, $0 < f < \hat{f}$, then we have $0 < \bar{e}_1 < \bar{e}'_1 < \bar{e}'_3 < \bar{e}_3 < e(R^e)$ such that:*

- (i) *when $\bar{e} \in (0, \bar{e}_1)$, $\vec{a}_1 = (c(R^c), 0, 0)$, $\vec{a}_2 = (0, d(R_0^d), 0)$, $q_1 = q_1^c$, $q_2 = q_2^d(R_0^d)$, and $R^d = R_0^d$;*
- (ii) *when $\bar{e} = \bar{e}_1$, a fraction $\lambda_1(1 - \lambda_1^e)$ of households choose $\vec{a}_1 = (c(R^c), 0, 0)$, a fraction $\lambda_1\lambda_1^e$ of households choose $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, $\vec{a}_2 = (0, d(R_0^d), 0)$, $q_1 = q_1^c$, $q_2 = q_2^d(R_0^d)$, and $R^d = R_0^d$;*
- (iii) *when $\bar{e} \in (\bar{e}_1, \bar{e}'_1]$, $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, $\vec{a}_2 = (0, d(R_0^d), 0)$, $q_1 = q_1^c$, $q_2 = q_2^d(R_0^d)$, and $R^d = R_0^d$;*
- (iv) *when $\bar{e} \in (\bar{e}'_1, \bar{e}'_3)$, $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, a fraction $\lambda_2\lambda_2^d$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$, a fraction $\lambda_2(1 - \lambda_2^d)$ of households choose $\vec{a}_2 = (0, \tilde{d}, \bar{e})$, $q_1 = q_1^c$, $q_2 = q_2^d(R^d) > q_2^d(R_0^d)$, and $R^d = \frac{R^e\bar{e}}{\delta + \bar{e}} > R_0^d$;*
- (v) *when $\bar{e} = \bar{e}'_3$, $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, a fraction $\lambda_2\lambda_2^d$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$ with DM consumption $q_2^d(R^d)$, a fraction $\lambda_2\lambda_2^{mix}$ of households choose $\vec{a}_2 = (0, \tilde{d}, \bar{e})$ with $q_2^d(R^d)$, a fraction $\lambda_2(\lambda_2^e - \lambda_2^{mix})$ of households choose $\vec{a}_2 = (0, 0, \bar{e})$ with $q_2^{\bar{e}}$, $q_1 = q_1^c$, and $R^d = \frac{R^e\bar{e}'_3}{\delta + \bar{e}'_3} > R_0^d$;*
- (vi) *when $\bar{e} \in (\bar{e}'_3, \bar{e}_3]$, $\vec{a}_1 = (\tilde{c}, 0, \bar{e})$, a fraction $\lambda_2\lambda_2^d$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$ with DM consumption $q_2^d(R^d)$, a fraction $\lambda_2(1 - \lambda_2^d)$ of households choose $\vec{a}_2 = (0, 0, \bar{e})$ with $q_2^{\bar{e}}$, $q_1 = q_1^c$, and R^d is such that $W_2^d(R^d) = W_2^{\bar{e}}$ with $R^d > R_0^d$;*
- (vii) *when $\bar{e} \in (\bar{e}_3, e(R^e)]$, $\vec{a}_1 = (0, 0, \bar{e})$, a fraction $\lambda_2\lambda_2^d$ of households choose $\vec{a}_2 = (0, d(R^d), 0)$ with DM consumption $q_2^d(R^d)$, a fraction $\lambda_2(1 - \lambda_2^d)$ of households choose $\vec{a}_2 = (0, 0, \bar{e})$ with $q_2^{\bar{e}}$, $q_1 = q_1^{\bar{e}} > q_1^c$, and R^d is such that $W_2^d(R^d) = W_2^{\bar{e}}$ with $R^d > R_0^d$,*

where $\bar{e}_3(\hat{R}^c) = \bar{e}'_3$.

Proof. See the Appendix.

Figure 6 presents the results stated in Proposition 7, showing how equilibrium portfolios evolve as the CBDC holding limit increases, when R^e is high, and both f and δ are not too large. Table 1 summarizes the comparative statics of the key variables with respect to an increase in the CBDC holding limit in each of the regions defined by the cutoffs.

Note three things here. One is that the CBDC is firstly adopted in Type 1 meetings, then in Type 2 meetings—when $R_0^d > R^c$ —as the marginal saving in the cost of accumulating

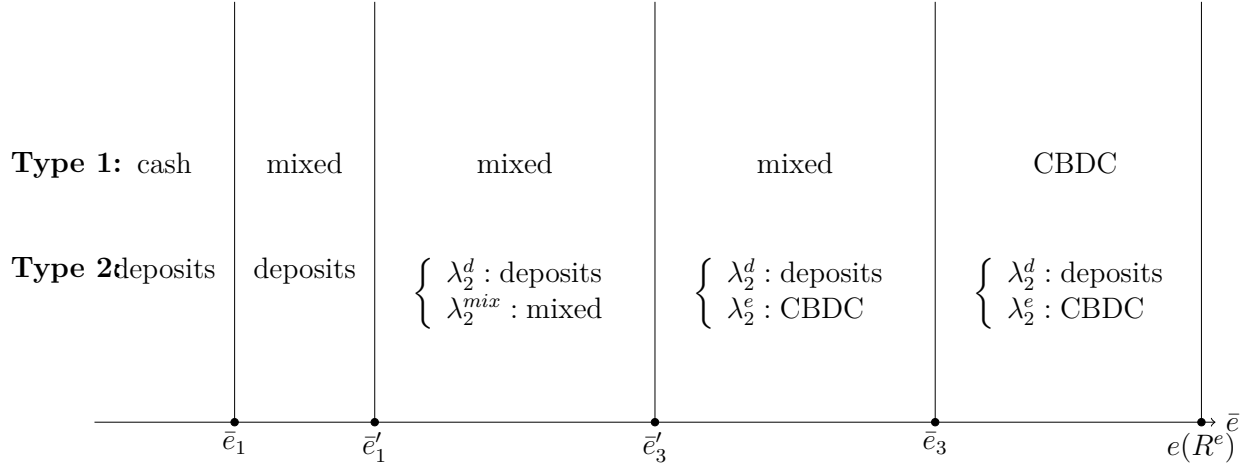


Figure 6: Optimal portfolios.

Notes. This figure shows the equilibrium portfolios under different ranges of $\bar{e}$ when $R^e > \max\{R^c, R_0^d\}$, $R_0^d > R^c > \hat{R}^c$, $W_2^d(R_0^d, f=0) > W_2^{\bar{e}=\bar{e}'_1}$, $0 < \delta < \hat{\delta}$, $0 < f < \hat{f}$.

Table 1: Comparative statics

	$\tilde{c}$	q_1	q_2^d	q_2^{mix}	$q_2^{\bar{e}}$	R^d	$\hat{\gamma}$	λ_2^d	λ_2^{mix}	λ_2^e
$[\bar{e}_1, \bar{e}'_1)$	$\downarrow$	—	—	—	—	—	—	—	—	—
$[\bar{e}'_1, \bar{e}'_3)$	—	—	$\uparrow$	$\uparrow$	0	$\uparrow$	$\uparrow$	$\downarrow$	$\uparrow$	0
$[\bar{e}'_3, \bar{e}_3)$	—	—	$\uparrow$	0	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	0	$\uparrow$
$[\bar{e}_3, e(R^e)]$	0	$\uparrow$	$\uparrow$	0	$\uparrow$	$\uparrow$	$\uparrow$	$\downarrow$	0	$\uparrow$

Notes. This table shows the comparative statics under different ranges of $\bar{e}$ when $R^e > \max\{R^c, R_0^d\}$, $R_0^d > R^c > \hat{R}^c$, $W_2^d(R_0^d, f=0) > W_2^{\bar{e}=\bar{e}'_1}$, $0 < \delta < \hat{\delta}$, $0 < f < \hat{f}$.

money is greater for buyers in Type 1 meetings when using the CBDC. Second, when $\bar{e} \in (\bar{e}_1, \bar{e}_3)$, $\bar{e}$ does not affect the amount of liquidity and hence DM consumption in Type 1 meetings where buyers hold a mix of cash and the CBDC. This is because after accumulating the CBDC to the limit, buyers continue topping up with cash until the marginal benefit of liquidity equals the marginal cost of holding cash. Hence, $\tilde{c} = \frac{q_1 - R^e \bar{e}}{R^c}$ adjusts to guarantee that the liquidity provided by cash fills the gap between payment required to trade q_1 and the liquidity provided by the CBDC. When the liquidity provided by the CBDC increases (decreases), the liquidity provided by cash decreases (increases). In other words, in this regime, changes in CBDC holding limits are neutral. This result aligns with the concept of credit neutrality discussed in Gu et al. (2016), in which real money balances adjust to changes in the credit limit so that the total amount of liquidity remains unchanged. Last but not least, after the CBDC is adopted in Type 2 meetings when $\bar{e} > \bar{e}'_1$, a larger $\bar{e}$ crowds out deposits and investment whereas a smaller $\bar{e}$ crowds in deposits and investment. Therefore, putting a holding limit on CBDC can help mitigate the crowding-out effect of the CBDC on bank deposits and investment.

5 CBDC Design

For discussions of optimal designs of the CBDC, welfare is defined as the sum of the utilities of all agents. Aggregate welfare is then

$$\begin{aligned} \mathcal{W} = & 2U(X^*) - H + \lambda_1 [\beta(u(q_1) - q_1)] \\ & + \lambda_2 \{(\lambda_2^d + \lambda_2^{mix}) [\beta(u(q_2^d(R^d)) - q_2^d(R^d))] + (\lambda_2^e - \lambda_2^{mix}) [\beta(u(q_2^e) - q_2^e)]\}, \end{aligned} \quad (30)$$

where H is aggregate working hours, which is obtained from the CM goods market-clearing condition

$$2X^* + \lambda_1 \lambda_1^e \delta_1 + \lambda_2 [(\lambda_2^d + \lambda_2^{mix})f + \lambda_2^e \delta_1] + \eta(\bar{\gamma} - \hat{\gamma}) = H + \eta \int_{\hat{\gamma}}^{\bar{\gamma}} \gamma d\gamma, \quad (31)$$

where the left-hand side is expenditure in CM goods, including households' optimal CM consumption, aggregate user costs for using deposits and the CBDC, and bankers' investment; the right-hand side is output from labor and bankers' project investment.

Substituting H obtained from (31) into (30), aggregate welfare then takes the following

expression

$$\begin{aligned}
\mathcal{W} = & 2(U(X^*) - X^*) + \eta \int_{\hat{\gamma}(R^d)}^{\bar{\gamma}} \left(\gamma - \frac{1}{\beta} \right) d\gamma + \lambda_1 [\beta(u(q_1) - q_1) - \lambda_1^e \delta_1] \\
& + \lambda_2 \left\{ \lambda_2^d [\beta(u(q_2^d(R^d)) - q_2^d(R^d)) - f] + \lambda_2^{mix} [\beta(u(q_2^d(R^d)) - q_2^d(R^d)) - f - \delta_1] \right. \\
& \left. + (\lambda_2^e - \lambda_2^{mix}) [\beta(u(q_2^e) - q_2^e) - \delta_1] \right\}. \tag{32}
\end{aligned}$$

In Section 5.1, I discuss properties of the welfare function when the CBDC holding limit is nonbinding, and in Section 5.2 when binding. Quantitative analysis on optimal CBDC designs is discussed in Section 5.3

5.1 Non-binding CBDC holding limit

As $\bar{e} \geq e(R^e)$, the central bank can only affect equilibrium allocations and thereby welfare through varying R^e and δ . Note here that the central bank can make the CBDC more (less) costly to adopt by setting a higher (lower) service charge fee, δ_2 . The CBDC user cost— δ_1 —cannot be directly adjusted by the central bank, as it depends on user-specific fixed effects. On one hand, a higher R^e and a lower δ will make the CBDC more likely to be adopted. On the other hand, given that the CBDC is adopted in both types of meetings, R^e and δ can have welfare impact with different magnitudes, and in general R^e is a better policy tool. Specifically, a higher R^e will lead to more efficient DM exchange across the board (i.e., q_1, q_2^d, q_2^e increase) while at the same time crowding out bank deposits and investment; a lower δ results in higher DM trade only in Type 2 meetings where only deposits are used (i.e., only q_2^d increases) while at the same time disintermediating private banks. Hence, if an increase in R^e and a decrease in δ cause the same extent of bank disintermediation, the former policy leads to a higher welfare than the latter. This result provides a justification for an interest-bearing CBDC and is formally recorded in the following proposition.

Proposition 8. *When $\bar{e} \geq e(R^e)$, $R^e > \max \{R^e(R^c, \delta), R^e(R_0^d, \delta)\}$, if an increase in R^e ($\Delta R^e > 0$) and a decrease in δ ($\Delta \delta < 0$) produce the same change in R^d ($\Delta R^d > 0$), then the welfare gain (loss) from the increase in R^e is larger (smaller) than that from the decrease in δ :*

$$\Delta \mathcal{W}(\Delta R^e) > \Delta \mathcal{W}(\Delta \delta).$$

In addition, changes in R^e affect aggregate welfare in a continuous way except when $R^e = R^e(R^c, \delta)$. When $R^e = R^e(R^c, \delta)$, buyers in Type 1 meetings are indifferent between

holding only cash and holding only the CBDC when they make their portfolio decisions in the CM and at this point they take the lump-sum transfer τ as given. Although compared with holding cash, holding the CBDC promotes more efficient DM exchange, the increase in DM consumption is exactly offset by the increase in CM working hours to pay the CBDC fixed cost, and this creates households' indifference between $\vec{a}_1^c = (c(R^c), 0, 0)$ and $\vec{a}_1^e = (0, 0, e(R^e))$. However, welfare is different for the two types of equilibria when $\lambda_1^c = 1, \lambda_1^e = 0$, and when $\lambda_1^c = 0, \lambda_1^e = 1$ as the lump-sum transfer from the central bank is different. This is because when the aggregate demand—hence the supply—for different types of central bank monies changes, central bank's seigniorage revenue changes, and hence the lump-sum transfer to households. The variation in lump-sum transfers across the two equilibria prevents changes in CM labor from perfectly absorbing changes in DM consumption. Therefore, welfare is different to the extent that τ is different. Let τ^c and τ^e denote the lump-sum transfers when $\lambda_1^c = 1, \lambda_1^e = 0$, and when $\lambda_1^c = 0, \lambda_1^e = 1$ at $R^e = R^e(R^c, \delta)$ respectively. The following proposition formalizes this discussion.

Proposition 9. *When $\bar{e} \geq e(R^e)$, at $R^e = R^e(R^c, \delta)$, $W_1^c = W_1^e$, but $\mathcal{W}(\lambda_1^c = 0, \lambda_1^e = 1) - \mathcal{W}(\lambda_1^c = 1, \lambda_1^e = 0) = \lambda_1 ((1 - \beta R^e) e(R^e) - (1 - \beta R^c) c(R^c) + \delta_2) \approx \tau^e - \tau^c$.*

5.2 Binding CBDC holding limit

In this section, I focus on the welfare impact of the CBDC holding limit $\bar{e}$. In what follows, I discuss how aggregate welfare behaves at two critical cutoffs: $\bar{e}_1$ and $\bar{e}'_3$.

First, when $R^e > R^c$ and $0 < \delta < \hat{\delta}$, at $\bar{e}_1$, buyers in Type 1 meetings are indifferent between holding only cash and holding a mix of cash and the CBDC, as $W_1^{mix}|_{\bar{e}=\bar{e}_1} = W_1^c$, where the CBDC adoption cost δ is exactly offset by savings in labor to acquire money when households take the lump-sum transfer as given. However, for the two types of equilibria when $\lambda_1^c = \lambda_1^e = 1$ and when $\lambda_1^c = 1, \lambda_1^e = 0$, the quantity and composition of real money balances differ. Consequently, the central bank's seigniorage revenue, and thus the lump-sum transfer to buyers, also differs. Let $\tilde{c}(\bar{e}_1)$ denote the real balance of cash held in the mixed portfolio with $\bar{e}_1$ units of the CBDC. We have $\tilde{c}(\bar{e}_1) + \bar{e}_1 < c(R^c)$. Denote τ^{mix} as the lump-sum transfer when $\lambda_1^c = \lambda_1^e = 1$. Proposition 10 shows how welfare changes when buyers in Type 1 meetings switch from holding only cash to holding both cash and the CBDC at $\bar{e}_1$. As it turns out, aggregate welfare changes by the amount of user costs incurred for adopting the CBDC.

Proposition 10. *When $\bar{e} < e(R^e)$, $R^e > R^c$, and $0 < \delta < \hat{\delta}$, at $\bar{e}_1$, $W_1^c = W_1^{mix}$, but $\mathcal{W}(\lambda_1^c = \lambda_1^e = 1) - \mathcal{W}(\lambda_1^c = 1, \lambda_1^e = 0) = \tau^{mix} - \tau^c = -\lambda_1 \delta_1$.*

Second, when $R^e > R_0^d$ and $0 < f < \hat{f}$, at $\bar{e}_3'$, households in Type 2 meetings are indifferent among the three types of portfolios: $(0, d(R^d), 0)$, $(0, \tilde{d}, \bar{e}_3')$, and $(0, 0, \bar{e}_3')$. However, aggregate welfare is different when $\lambda_2^{mix} = \lambda_2^e > 0$ where the CBDC is held alongside deposits at the intensive margin and when $\lambda_2^{mix} = 0$ where the CBDC is held alongside deposits only at the extensive margin. As the equilibrium switches from the former to the latter, since deposits no longer coexist with the CBDC at the intensive margin, λ_2^d increases to ensure that aggregate demand for deposits remain unchanged. As a consequence, the fraction of buyers holding the CBDC decreases, meaning that real balances of the CBDC drop. Thus, the lump-sum transfer from the central bank differs across the two equilibria. Denote λ_2^{md} as the fraction of buyers switching from holding the mixed portfolio to holding only deposits. Proposition 11 shows how welfare responds when the equilibrium changes from $\lambda_2^{mix} = \lambda_2^e > 0$ to $\lambda_2^{mix} = 0$ at $\bar{e}_3'$.

Proposition 11. *When $\bar{e} < e(R^e)$, $R^e > R_0^d$, and $0 < f < \hat{f}$, at $\bar{e}_3'$, $W_2^d(R^d) = W_2^{\bar{e}} = W_2^{mix}(R^d)$, but $\mathcal{W}(\lambda_2^{mix} = \lambda_2^e > 0) - \mathcal{W}(\lambda_2^{mix} = 0) = \lambda_2 \lambda_2^{md} ((1 - \beta R^e) \bar{e} + \delta_2)$.*

5.3 Quantitative analysis

In this section, I use data on the United States economy to first calibrate the model parameters without a CBDC, and then conduct counterfactual analysis to study the optimal design of a CBDC.

Calibration. I modify the model in two aspects. First, I introduce search frictions in the DM so the probability that a buyer meets a seller is $\alpha \in [0, 1]$. Second, the terms of trade in the DM is determined by the proportional Kalai bargaining, with sellers' share of the total trade surplus being $1 - \theta$. This captures the fact that sellers enjoy substantial markups in the data. The two modifications do not affect the results qualitatively, but can matter quantitatively.

Consider an annual model. CM and DM utility functions take the logarithmic form $U(X) = A \log(X)$ and CRRA form $u(q) = \frac{(q+B)^{1-\sigma} - B^{1-\sigma}}{1-\sigma}$ respectively. The parameter B is set to 0.001, which ensures that $u(0) = 0$. The measure of bankers, η , is normalized to one without loss of generality. There are 11 parameters to calibrate: $(\beta, \lambda_1, \lambda_2, \mu, \epsilon, \alpha, A, \sigma, \theta, \bar{\gamma}, f)$. There is a direct match for the first five parameters: $(\beta, \lambda_1, \lambda_2, \mu, \epsilon)$. The rest has to be calibrated internally. Parameters are primarily calibrated using data from 1987-2008; for data not available in this range, observations from alternative years are used.

The data used in my calibration exercise come from six sources: (1) data from the Survey of Consumer Payment Choice (SCPC) and the Diary of Consumer Payment Choice

(DCPC) from the Federal Reserve Bank of Atlanta; (2) data from the survey of costs of payments in Denmark; (3) haircuts applied to high-quality liquid assets (HQLA) in the Basel Framework; (4) new M1 series from Lucas and Nicolini (2015); (5) call report data from the Federal Financial Institutions Examination Council; (6) several time series on macro variables, reserves, and retail value added from Federal Reserve Economic Data (FRED). The calibration of several key parameters is discussed briefly below.

I obtain the fractions of Type 1 and Type 2 meetings, λ_1 and λ_2 , from the SCPC (Greene and Stavins 2018) and the DCPC (Premo, 2018). From the SCPC, I estimate the fraction of transactions conducted online, and the DCPC provides information on the perceived percentage of transactions that reject cash or debit/credit card payments at the point of sale. I use data from surveys in 2016, and the numbers are similar in 2015 and 2017. In the context of my model, λ_1 is the fraction of transactions that accept only cash, which are point-of-sale transactions where cards are not accepted, and λ_2 is the fraction that accepts only deposits, including both online transactions and point-of-sale transactions where cash is not accepted. The fraction of online transactions is estimated at 25.37% from the SCPC. According to the DCPC, at the point of sale, 6.14% of transactions do not accept debit or credit cards, while 1.44% do not accept cash. Hence, the fraction of cash-only transactions is $6.14\% \times (1 - 25.37\%) = 4.58\%$, and the fraction of deposit-only transactions is $25.37\% + 1.44\% \times (1 - 25.37\%) = 26.44\%$. With normalization, $\lambda_1 = \frac{4.58}{4.58+26.44} = 14.76\%$, and $\lambda_2 \equiv 1 - \lambda_1 = 85.24\%$.

Next, I calibrate the DM trading probability (α), the two utility function parameters (A, σ), and the bargaining power (θ) jointly to match the money demand curve, the retail value-added share of output, and a 20% retail markup. I use the new M1 series from Lucas and Nicolini (2015) for the monetary aggregate. The deposit rate used to calculate the money demand in the model is obtained from the interest expenses and balances on transactional deposits series in the call report data, which was also used in Drechsler et al. (2017). The retail value-added share of output takes the average from 2005 to 2008, since data on retail value-added is not available prior to 2005. However, it is reasonable to assume that this share does not fluctuate significantly over the 1987–2008 period, as it has remained stable from 2005 to 2024.

To calibrate f , the deposit user cost, I construct an auxiliary variable, $\hat{f}$, the per period service charge on deposit accounts. There is no service charge on deposit account in the model, as I assume perfect competition, however, in the data there is, as it is not perfect competition. I set $\hat{f}$ to match service charges per dollar on deposit accounts calculated using series from the call report data.¹⁰ Then, f is chosen to match the ratio between

¹⁰Service charges on deposit accounts, among other components, include periodic maintenance fees. De-

the time-related costs of checking account statements and the annual subscription fees for international debit cards in Denmark in 2009. The ratio is $\frac{19.6}{45} = 0.44$. Hence, $f = 0.44 \times \hat{f}$.

Table 2 lists all calibrated parameter values with their targets.

Table 2: Calibration Results

Parameters	Notation	Value	Calibration Targets
<i>Calibrated externally</i>			
Discount factor	β	0.96	Standard in literature
Reserve requirement	μ	2.4%	1987-2008 avg. required reserves/trans. balances
Bank's pledgeability constraint	ϵ	0.85	Haircut applied to level 2A assets in Basel III
Frac. of Type 1 meetings	λ_1	14.76%	SCPC2016, DCPC2016
Frac. of Type 2 meetings	λ_2	85.24%	SCPC2016, DCPC2016
<i>Calibrated internally</i>			
Prob. of DM trading	α	0.51	Retail value-added 2005-08
Coeff. on CM consumption	A	3.57	Money demand 1987-2008
Curv. of DM consumption	σ	0.42	Money demand 1987-2008
Buyer's bargaining power	θ	0.76	Retailer markup 20%
Upper bound of bank's productivity	$\bar{\gamma}$	1.87	1987-2008 avg. interest rate on transactional deposits
Resource cost of deposits	f	0.0024	Resource cost/subscription fee 2009

Effects of a CBDC holding limit. I now introduce a universal CBDC that pays twice the net nominal interest rate of deposits, has the same fixed cost as deposits with no service charge, and conduct counterfactual analysis. I am particularly interested in how the CBDC holding limits affect equilibrium allocations, investment, and welfare. Figure 7 shows the results. In all figures, the horizontal axis represents the CBDC holding limit in real terms, which is bounded above by $e(R^e)$. The first row displays the quantity of DM trade in Type 1 and Type 2 meetings, and the fractions of households holding only deposits, the mixture, and only the CBDC in Type 2 meetings. The second row shows the amount of deposits in the mixed portfolio, the equilibrium gross real deposit rate, and aggregate welfare.

Once CBDC is adopted, a higher $\bar{e}$ leads to more efficient DM trade while at the same time crowds out deposits and investment. q_1 and $q_2^{\bar{e}}$ respond more sharply to $\bar{e}$ than q_2^d and q_2^{mix} , as the former two are directly affected by the limit while the latter two are indirectly affected through changes in the deposit rate. The welfare function is not differentiable at

posit accounts consist of both transactional and non-transactional accounts in the call report data, where only the former is discussed in the paper. However, in the model, it is the relative size of the fixed costs of bank deposits and CBDC that affects households' portfolio decisions; therefore, this caveat does not undermine the quantitative results.

the cutoffs where the equilibrium changes and it reaches its maximum when $\bar{e}$ hits the upper bound, implying the gains in DM exchange efficiency outweighs the crowding out on investment as $\bar{e}$ increases. Note that there are two welfare drops: one when the CBDC is adopted in Type 1 meetings at $\bar{e}_1$; another when buyers in Type 2 meetings stop holding a mix of deposits and the CBDC at $\bar{e}'_3$. In the first case, welfare drops by $\lambda_1 \delta_1$ —the fixed cost for adopting the CBDC; in the second case, the amount of liquidity falls, and hence the drop in welfare.

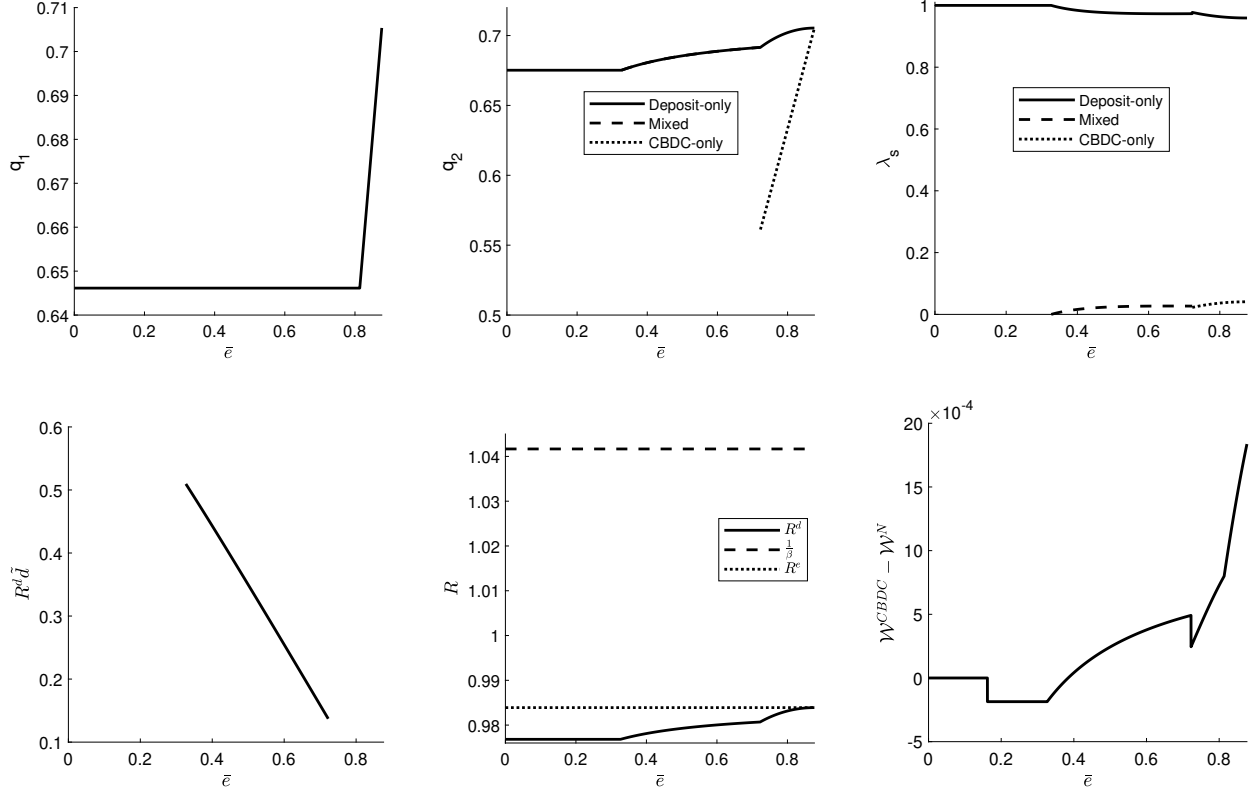


Figure 7: Effects of the CBDC holding limit.

Deposit-like CBDC. In this part, I discuss the optimal design of a deposit-like CBDC, which competes only with bank deposits in Type 2 meetings. A deposit-like CBDC lacks the offline functionality which is necessary for use in Type 1 meetings. I show several things. First, when the CBDC interest rate is not set optimally, a binding CBDC holding limit can be desirable as it improves welfare. Nonetheless, the optimal limit is nonbinding at the optimal CBDC interest rate. In other words, we do not need the limit when the interest rate feature works perfectly well, but this is not the case when we fail to set the interest rate optimally. Figure 8 shows the welfare gain (loss) against the CBDC limit when $\epsilon = 0.77$, $i = 2 \times i_d^0$ and $\delta = f$, where i_d^0 is the benchmark net nominal deposit rate. The plot on the

right-hand side zooms in on the one on the left-hand side, stopping at $\bar{e}'_3$ before households become indifferent between holding only the CBDC and holding only bank deposits. In this scenario, it is optimal to impose a cap on CBDC holdings at which households are indifferent between holding only deposits and holding a mix of deposits and the CBDC. Furthermore, Figure 9 shows the optimal interest rate and limit when $\epsilon = 0.77$ and $\delta = f$. The left panel plots the welfare gain (loss) against the CBDC interest rate when we have the optimal limit at each rate. The black solid line in the right panel plots the optimal limit for each CBDC interest rate, and the red dashed line is the upper bound of a binding CBDC limit. The optimal interest rate is around 1.4 times the benchmark net nominal deposit rate, and the optimal limit is nonbinding at this interest rate.

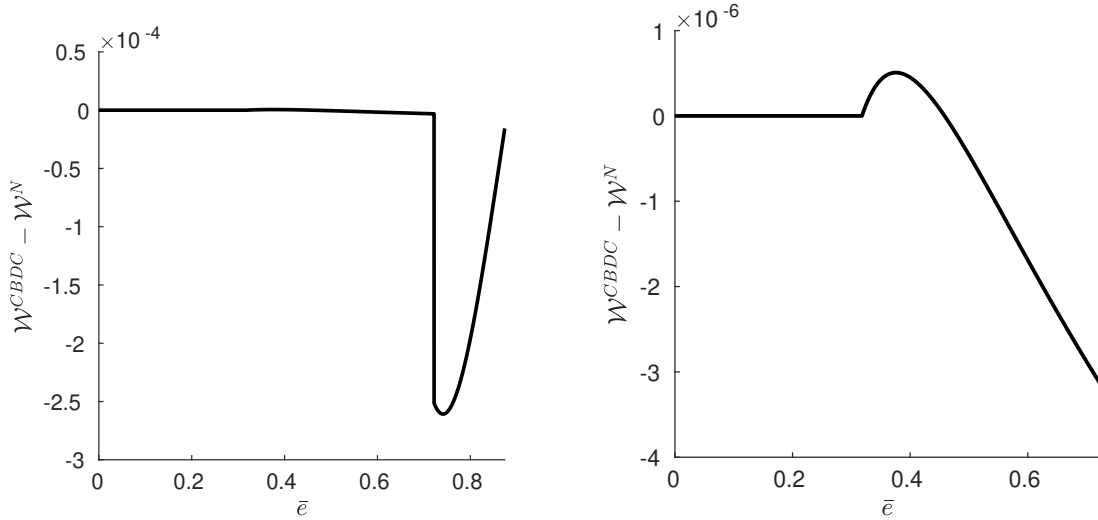


Figure 8: Optimal limit with a deposit-like CBDC.

Notes. This figure shows the welfare gain (loss) against the deposit-like CBDC limit when $\epsilon = 0.77$, $i = 2 \times i_d^0$ and $\delta = f$, where i_d^0 is the benchmark net nominal deposit rate.

Second, welfare increases as δ increases. The rationale is that a higher δ leads to higher R^d and R^e and this positive DM effect dominates the negative effect in investment. However, in practice, it may be difficult for the central bank to collect higher CBDC service charges δ_2 from households as people can try to avoid the service charge the way they avoid taxes. Hence, I impose the following cost function

$$f(\delta_2) = \frac{\chi}{2} \max\{0, \delta_2\}^2, \quad (33)$$

where $\chi > 0$ controls the convexity of the cost function. Figure 10 shows the welfare gain (loss) against δ when we have the optimal interest rate and limit at each δ . In the left panel, setting $\delta_2 > 0$ entails no cost whereas in the right panel increasing δ_2 is subject to a cost

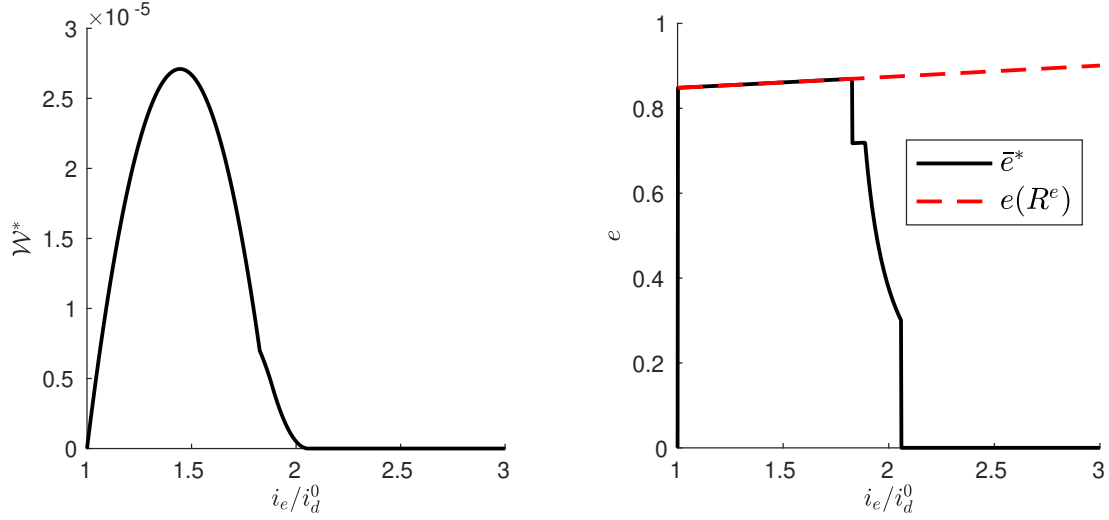


Figure 9: Optimal interest rate and limit with a deposit-like CBDC.

Notes. This figure shows the optimal interest rate and limit of a deposit-like CBDC when $\epsilon = 0.77$ and $\delta = f$.

with $\chi = 2$.

6 Conclusion

The introduction of a CBDC can crowd out bank deposits. While policymakers in some countries have come up with the idea of imposing a cap on households' CBDC wallet so as to mitigate its disruptions to private banks, the equilibrium implications of such a cap and its interaction with other CBDC-related policy tools remain largely unexplored.

My analysis shows how a relatively standard framework of banking and payment within the New Monetarist tradition can shed light on these issues. The findings suggest that imposing a binding CBDC holding limit may or may not be optimal depending on the interest rate and fixed cost of CBDC. At the optimal interest rate, we do not need a limit, however, when the central bank fails to set the CBDC interest rate optimally, a limit improves welfare.

The CBDC design proposed in this paper can inform the regulation of stablecoins. If stablecoins were to gain widespread use as a means of payment, they could crowd out bank deposits too; therefore a cap on stablecoins could prove useful in this context.

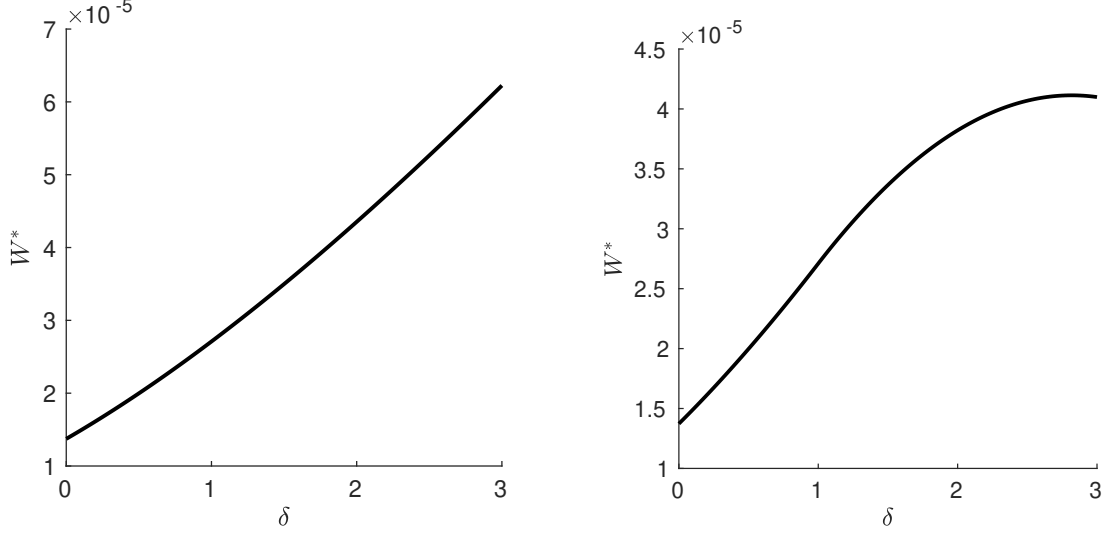


Figure 10: Optimal δ with a deposit-like CBDC.

Notes. This figure shows the optimal design of a deposit-like CBDC when $\epsilon = 0.77$.

Appendix A Proofs of Propositions and Lemmas

A.1 Proof of Terms of Trade

Proof. The sellers' participation constraint (2) must hold at equality since otherwise buyers could increase their surplus by slightly reducing the amount they offer to pay sellers so that sellers would still find the offer acceptable. In addition, the constraint, $0 \leq \vec{p}_j$, is never binding because autarky is never optimal. Thus, the terms of trade solves the following problem

$$\max_{q_j, \vec{p}_j} u(q_j) - \vec{R} \cdot \vec{p}_j \quad \text{s.t.} \quad -q_j + \vec{R} \cdot \vec{p}_j = 0, \vec{p}_j \leq \vec{a}_j$$

$\Leftrightarrow \max_{q_j, \vec{p}_j} u(q_j) - q_j \quad \text{s.t.} \quad \vec{p}_j \leq \vec{a}_j$. When $\vec{p}_j < \vec{a}_j$, $u'(q_j) - 1 = 0$ pins down q_j , which is q^* ; when $\vec{p}_j = \vec{a}_j$, $q_j = \vec{R} \cdot \vec{a}_j$. $\square$

A.2 Proof of Lemma 1

Proof. For buyers in Type 1 meetings to hold both cash and the CBDC, the following first-order conditions for cash and CBDC holdings will hold with equality if and only if $R^c = R^e$.

$$\begin{aligned}\Lambda\left(\vec{R} \cdot f_1(\vec{a})\right) &\leq \frac{1}{\beta R^c} - 1, \\ \Lambda\left(\vec{R} \cdot f_1(\vec{a})\right) &\leq \frac{1}{\beta R^e} - 1.\end{aligned}$$

However, when $R^c = R^e \Rightarrow W_1^c > W_1^e$, as $\delta > 0$. So buyers will only hold cash. $\square$

A.3 Proof of Proposition 2

Proof. By setting $W_1^c = W_1^e \Leftrightarrow (\beta R^c - 1)c(R^c) + \beta(u(q_1^c) - q_1^c) + \Omega_1 = (\beta R^e - 1)e(R^e) + \beta(u(q_1^e) - q_1^e) - \delta + \Omega_1 \Leftrightarrow (\beta R^c - 1)c(R^c) + \beta(u(R^c c(R^c)) - R^c c(R^c)) = (\beta R^e - 1)e(R^e) + \beta(u(R^e e(R^e)) - R^e e(R^e)) - \delta \Rightarrow R^e(R^c, \delta)$. $\square$

A.4 Proof of Lemma 2

Proof. For buyers in Type 2 meetings to hold both deposits and the CBDC, the following first-order conditions for deposits and CBDC holdings will hold with equality if and only if $R^d = R^e$

$$\begin{aligned}\Lambda\left(\vec{R} \cdot f_2(\vec{a})\right) &\leq \frac{1}{\beta R^d} - 1, \\ \Lambda\left(\vec{R} \cdot f_2(\vec{a})\right) &\leq \frac{1}{\beta R^e} - 1.\end{aligned}$$

However, as $R^d = R^e$ and $f + \delta > \max\{f, \delta\}$, buyers obtain the same amount of DM surplus subject to the same marginal cost of holding money when holding both deposits and the CBDC compared with when holding only one of them, while paying two fixed costs. So, buyers will choose to hold only one type of money. $\square$

A.5 Proof of Proposition 3

Proof. By setting $W_2^d(R_0^d) = W_2^e \Leftrightarrow R^e(R_0^d, \delta)$. When $R^e > R^e(R_0^d, \delta)$, the deposit rate R_0^d will increase to R^d until $W_2^d(R^d) = W_2^e$. The fraction of buyers holding deposits, $\lambda_2^d(R^d)$, is pinned down by the deposit market clearing condition: $\lambda_2^d = \frac{\eta(\bar{\gamma} - \hat{\gamma}(R^d))}{\lambda_2 d(R^d)}$. $\bar{\gamma} - \hat{\gamma}(R^d(R^e)) = 0 \Rightarrow \bar{\gamma}(R^e)$, and when $\bar{\gamma} > \bar{\gamma}(R^e)$, $\lambda_2^d > 0$. $\square$

A.6 Proof of Proposition 4

Proof. 1. $R^e \uparrow \Rightarrow e(R^e) \uparrow \Rightarrow q_1^e, q_2^e \uparrow$;

2. $R^e \uparrow \Rightarrow W_2^e \uparrow \Rightarrow R^d \uparrow$ so that $W_2^d(R^d) = W_2^e \Rightarrow q_2^d \uparrow, \lambda_2^d = \frac{\eta(\bar{\gamma} - \hat{\gamma}(R^d))}{\lambda_2 d(R^d)} \downarrow$.

□

A.7 Proof of Lemma 3

Proof. 1. When $R^e, \bar{e}$ are low, and δ is large, buyers will want to use cash;

2. when $R^e, \bar{e}$ are high, and δ is small, buyers will want to use the CBDC;

3. when R^e is high, $\bar{e}, \delta$ are low, buyers will want to use both cash and the CBDC such that $\Lambda(R^e \bar{e} + R^c \tilde{c}) = \frac{1}{\beta R^c} - 1 = \Lambda(R^c c(R^e))$.

□

A.8 Proof of Lemma 4

Proof. When holding only $\bar{e}$ units of the CBDC, the FOC is $\Lambda(R^e \bar{e}) < \frac{1}{\beta R^e} - 1 < \frac{1}{\beta R^c} - 1$, so it is optimal to top up with cash until $\Lambda(R^e \bar{e} + R^c \tilde{c}) = \frac{1}{\beta R^c} - 1$.

□

A.9 Proof of Lemma 5

Proof. If $R^e > R^c$, then $\tilde{c} + \bar{e} = c(R^c) + (1 - \frac{R^e}{R^c})\bar{e} < c(R^c)$.

□

A.10 Proof of Lemma 6

Proof. 1. First, when $0 < \delta < \hat{\delta}$, $\bar{e}_1 < \bar{e}_3$;

2. Second, when $\bar{e}_1 < \bar{e}_3$, $W_1^{\bar{e}_1} < W_1^{mix}|_{\bar{e}=\bar{e}_1} = W_1^c = W_1^{\bar{e}_2} = W_1^c < W_1^{mix}|_{\bar{e}=\bar{e}_3} = W_1^{\bar{e}_3}$. $W_1^{\bar{e}}$ is strictly increasing in $\bar{e}$, so $\bar{e}_1 < \bar{e}_2 < \bar{e}_3$.

□

A.11 Proof of Proposition 5

Proof. 1. When one of the following conditions is true, $W_1^c > \max\{W_1^{\bar{e}}, W_1^{mix}\}$:

(i) $R^e \leq R^c$, or

(ii) $R^e > R^c$, $0 < \delta < \hat{\delta}$, $\bar{e} < \bar{e}_1$, or

- (iii) $R^e > R^c$, $\delta \geq \hat{\delta}$, $\bar{e} < \bar{e}_2$;
- 2. when $R^e > R^c$, $\bar{e} > \max\{\bar{e}_2, \bar{e}_3\}$, $W_1^{\bar{e}} > \max\{W_1^c, W_1^{mix}\}$;
- 3. when $R^e > R^c$, $0 < \delta < \hat{\delta}$, and $\bar{e}_1 < \bar{e} < \bar{e}_3$, $W_1^{mix} > \max\{W_1^c, W_1^{\bar{e}}\}$.

□

A.12 Proof of Lemma 7

Proof. If $R^e > R^d$, then $\tilde{d} + \bar{e} = d(R^d) + (1 - \frac{R^e}{R^d})\bar{e} < d(R^d)$.

□

A.13 Proof of Lemma 8

- Proof.* 1. $W_2^d(R_0^d, f) = W_2^{\bar{e}'_1} \Rightarrow \hat{f}$ such that $\bar{e}'_1 < \bar{e}'_2$ if $f < \hat{f}$, and $\bar{e}'_1 \geq \bar{e}'_2$ if $f \geq \hat{f}$. And when $W_2^d(R_0^d, f = 0) > W_2^{\bar{e}'_1}$, $\hat{f} > 0$;
2. $f < \hat{f} \Rightarrow \bar{e}'_1 < \bar{e}'_2 \Rightarrow W_2^{\bar{e}'_1} < W_2^{\bar{e}'_2} = W_2^d(R_0^d) = W_2^{mix}(R_0^d(\bar{e}'_1))$. Note $W_2^{\bar{e}'_3} = W_2^{mix}(R^d(\bar{e}'_3))$. With Assumption 3, we thus have $\bar{e}'_1 < \bar{e}'_3 \Rightarrow R^d(\bar{e}'_3) > R_0^d(\bar{e}'_1) \Rightarrow W_2^d(R_0^d) = W_2^{\bar{e}'_2} < W_2^{\bar{e}'_3} = W_2^{mix}(R^d(\bar{e}'_3)) = W_2^d(R^d(\bar{e}'_3)) \Rightarrow \bar{e}'_2 < \bar{e}'_3$;
3. $W_2^{\bar{e}'_4} > W_2^{mix}(R^d(\bar{e}'_4))$ if $f > 0$; $W_2^{\bar{e}'_3} = W_2^{mix}(R^d(\bar{e}'_3))$. With Assumption 3, we thus have $\bar{e}'_3 < \bar{e}'_4$;
4. Taken together, when $0 < f < \hat{f}$, we have $\bar{e}'_1 < \bar{e}'_2 < \bar{e}'_3 < \bar{e}'_4$.

□

A.14 Proof of Proposition 6

- Proof.* 1. When one of the following conditions is true, $W_2^d(R_0^d) > \max\{W_2^{\bar{e}}, W_2^{mix}(R^d)\}$:
- (i) $R^e \leq R_0^d$ and $\bar{e} \leq \bar{e}'_2$, or
 - (ii) $R^e > R_0^d$ and $\bar{e} \leq \min\{\bar{e}'_1, \bar{e}'_2\}$;
2. when $R^e > R_0^d$, $\bar{e}'_1 < \bar{e} < \bar{e}'_3$, $W_2^{mix}(R^d) > \max\{W_2^{\bar{e}}, W_2^d(R_0^d)\}$, so the deposit rate will increase until $W_2^{mix}(R^d) = W_2^d(R^d)$, where households are indifferent between holding only deposits or the mix;
3. when one of the following conditions is true, $W_2^{\bar{e}} > \max\{W_2^d(R_0^d), W_2^{mix}(R^d)\}$, so the deposit rate will increase until $W_2^d(R^d) = W_2^{\bar{e}}$, where households are indifferent between holding only deposits or only the CBDC:

- (i) $R^e \leq R_0^d$ and $\bar{e} > \bar{e}'_2$, or
- (ii) $R^e > R_0^d$ and $\bar{e} > \max\{\bar{e}'_2, \bar{e}'_3\}$.

□

A.15 Proof of Proposition 7

Proof. 1. $R_0^d > R^c \Rightarrow 0 < \bar{e}_1 < \bar{e}'_1$;

2. $R^c > \hat{R}^c \Rightarrow \bar{e}'_3 < \bar{e}_3$;

3. $R^e > R^c \Rightarrow \bar{e}_3 < e(R^e)$.

□

A.16 Proof of Proposition 8

Proof. Suppose the economy starts from an equilibrium where aggregate welfare is $\mathcal{W}_\infty$. Now consider two sets of policy:

- 1. R^e increases to $R^{e'}$ so that R^d increases by $\Delta(R^d)$. Aggregate welfare become $\mathcal{W}_2$;
- 2. δ decreases to δ' so that R^d increases by $\Delta(R^d)$ too. Aggregate welfare become $\mathcal{W}_3$.

We have

$$\mathcal{W}_2 - \mathcal{W}_3 = \lambda_1 \beta (\Delta u(q_1) - \Delta q_1) + \lambda_2 (1 - \lambda_2^d) \beta (\Delta u(q_2^e) - \Delta q_2^e),$$

where $\Delta q_1, \Delta q_2^e > 0 \Rightarrow \mathcal{W}_2 - \mathcal{W}_3 > 0$.

□

A.17 Proof of Proposition 9

Proof. First, from the aggregate welfare function (32), I have

$$\Delta \mathcal{W} = \lambda_1 (\beta \Delta u - \beta \Delta q - \delta_1). \quad (\text{A.1})$$

Substituting the following conditions into (A.1)

$$\begin{aligned} \beta \Delta u &= -c(R^c) + e(R^e) + \delta, \\ \beta \Delta q &= \beta (R^e e(R^e) - R^c c(R^c)), \end{aligned}$$

I obtain

$$\Delta \mathcal{W} = \lambda_1 ((1 - \beta R^e) e(R^e) - (1 - \beta R^c) c(R^c) + \delta_2).$$

Second, denote the price of money as ϕ_1 and ϕ_2 , and the aggregate supply of cash and CBDC as C_1, E_1 and C_2, E_2 for the equilibrium when $\lambda_1^c = \lambda_1$, $\lambda_1^e = 0$, and when $\lambda_1^c = 0$, $\lambda_1^e = \lambda_1$, respectively. Then I have

$$\Delta \tau = \phi_2 (C_2 + E_2 - C_{2,-1} - (1 + i)E_{2,-1}) - \phi_1 (C_1 + E_1 - C_{1,-1} - (1 + i)E_{1,-1}) + \lambda_1 \delta_2. \quad (\text{A.2})$$

Substituting the following market-clearing conditions into (A.2)

$$\begin{aligned} \phi_1 C_1 &= \lambda_1 c + \mu \lambda_2 \lambda_2^d d, \\ \phi_2 C_2 &= \mu \lambda_2 \lambda_2^d d, \\ \phi_1 E_1 &= \lambda_2 \lambda_2^e e(R^e), \\ \phi_2 E_2 &= (\lambda_1 + \lambda_2 \lambda_2^e) e(R^e), \end{aligned}$$

I obtain

$$\Delta \tau = \lambda_1 ((1 - R^e) e(R^e) - (1 - R^c) c(R^c) + \delta_2).$$

Hence, $\Delta \mathcal{W} \approx \Delta \tau$ with $\beta \approx 1$. □

A.18 Proof of Proposition 10

Proof. As the amount of DM trade, portfolios in Type 2 meetings and investment remain unchanged, change in aggregate welfare solely comes from changes in hours worked by households in Type 1 meetings.

$$\Delta \mathcal{W} = -\Delta h = -\lambda_1 (\tilde{c} + \bar{e} - c(R^c) + \delta) + \tau^{mix} - \tau^c. \quad (\text{A.3})$$

For the equilibrium where households in Type 1 meetings use only cash, denote the price of money, and aggregate supply of cash and CBDC as ϕ^c , C and E ; for the equilibrium where households in Type 1 meetings hold the mixed portfolio, denote the price of money, and aggregate supply of cash and CBDC as ϕ^{mix} , C^{mix} and E^{mix} . Then I have

$$\begin{aligned} \tau^c &= \phi^c (C + E) - \phi^c (C_{-1} + E_{-1}), \\ \tau^{mix} &= \phi^{mix} (C^{mix} + E^{mix}) - \phi^{mix} (C_{-1}^{mix} + (1 + i)E_{-1}^{mix}) + \lambda_1 \delta_2. \end{aligned}$$

Together with the following market-clearing conditions

$$\begin{aligned}\phi^c C &= \lambda_1 c(R^c) + \mu \lambda_2 (\lambda_2^d d + \lambda_2^{mix} \tilde{d}), \\ \phi^{mix} C^{mix} &= \lambda_1 \tilde{c} + \mu \lambda_2 (\lambda_2^d d + \lambda_2^{mix} \tilde{d}), \\ \phi E &= \lambda_2 \lambda_2^e \bar{e}, \\ \phi^{mix} E^{mix} &= (\lambda_1 + \lambda_2 \lambda_2^e) \bar{e},\end{aligned}$$

and the relationship $c(R^c) = \tilde{c} + (1 + i)\bar{e}$, I have

$$\tau^{mix} - \tau^c = \lambda_1 (\tilde{c} + \bar{e} - c(R^c) + \delta_2). \quad (\text{A.4})$$

Substituting (A.4) into (A.3), I arrive at $\Delta \mathcal{W} = -\lambda_1 \delta_1$. $\square$

A.19 Proof of Proposition 11

Proof. First, from the aggregate welfare function (32), I have

$$\Delta \mathcal{W} = \lambda_2 (\lambda_2^e (\beta \Delta u - \beta \Delta q + f) + \lambda_2^{md} \delta_1). \quad (\text{A.5})$$

Substituting the following equilibrium conditions into (A.5)

$$\begin{aligned}-\beta \Delta u &= \tilde{d} + f, \\ -\beta \Delta q &= \beta R^d \tilde{d}, \\ \lambda_2^e \tilde{d} &= \lambda_2^{md} (d - \tilde{d}) = \lambda_2^{md} (\delta + \bar{e}), \\ R^d &= \frac{R^e \bar{e}}{\delta + \bar{e}},\end{aligned}$$

I have

$$\Delta \mathcal{W} = -\lambda_2 \lambda_2^{md} ((1 - \beta R^e) \bar{e} + \delta_2).$$

Second, denote the price of money and aggregate supply of CBDC as ϕ_1 and E_1 for the equilibrium where $\lambda_2^{mix} = \lambda_2^e > 0$, and as ϕ_2 and E_2 for the equilibrium where $\lambda_2^{mix} = 0$. Then I have

$$\Delta \tau = \phi_2 (E_2 - (1 + i)E_{2,-1}) - \phi_1 (E_1 - (1 + i)E_{1,-1}) - \lambda_2 \lambda_2^{md} \delta_2. \quad (\text{A.6})$$

Substituting the following market-clearing conditions into (A.6)

$$\begin{aligned}\phi_1 E_1 &= (\lambda_1 \lambda_1^e + \lambda_2 \lambda_2^{mix}) \bar{e}, \\ \phi_2 E_2 &= (\lambda_1 \lambda_1^e + \lambda_2 \lambda_2^e) \bar{e},\end{aligned}$$

I obtain

$$\Delta\tau = -\lambda_2 \lambda_2^{md} ((1 - R^e) \bar{e} + \delta_2).$$

Hence, $\Delta\mathcal{W} \approx \Delta\tau$ with $\beta \approx 1$. □

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